



VIDYA VIKAS INSTITUTE OF ENGINEERING AND TECHNOLOGY, MYSORE

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VTU Previous Question Papers

**“Information Science and Engineering”
IV, VI & VIII Semester
Dec.2024/Jan.2025**

VVIET
Bannur road, Alanahalli, Mysore-570028

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4	BCS401	Analysis and Design of Algorithms	Dec.2024/Jan.2025
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Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Advanced Linear Algebra

Time: 3 hrs.

Max. Marks: 100

Note : Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the vector space with examples. (06 Marks)
 b. Test the consistency and solve them $x - 3y + 4z = -4$, $3x + 7y + 7z = -8$ and $-4x + 6y - z = 7$. (07 Marks)
 c. Solve the linear system of equations $3x + 2y + 7z = 4$, $2x + 3y + z = 5$, $3x + 4y + z = 7$ using LU - decomposition. (07 Marks)

OR

- 2 a. Define the terms : i) Subspace ii) Bases and iii) Dimensions. (06 Marks)
 b. Prove that intersection of two subspaces of a V.S $V(F)$ is a subspace of $V(F)$. (07 Marks)
 c. Show that the set $S = \{V_1, V_2, V_3, V_4\}$ where $V_1 = (1, 0, 1, 0)$, $V_2 = (0, 1, -1, 2)$, $V_3 = (0, 2, 2, 1)$, $V_4 = (1, 0, 0, 1)$. (07 Marks)

Module-2

- 3 a. Define the terms : i) Linear transformation ii) Rank iii) Nullity. (06 Marks)
 b. Let v and w be V.S $V(F)$ and let T be a L.T from v into w and let v be a finite dimension. Prove that $\text{rank}(T) + \text{nullity}(T) = \dim v$. (07 Marks)
 c. Show that the mapping $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x + 1, y, z)$ are linear transformation. (07 Marks)

OR

- 4 a. Let v and w be V.S. $V(F)$ and let $T : v \rightarrow w$. Prove that T is invertible and T^{-1} is also invertible. (06 Marks)
 b. Show that the vectors $V_1 = (1, 1, 0, 0)$, $V_2 = (2, 0, 1, 0)$, $V_3 = (0, 1, 2, -1)$, $V_4 = (0, 1, -1, 0)$ forms a basis for $\mathbb{R}^4$. Find the coordinates of each of the standard basis vectors in the ordered basis $\{V_1, V_2, V_3, V_4\}$. (07 Marks)
 c. Let $T : V \rightarrow W$ is a Linear transformations from the $V(F)$ to $W(F)$. Prove that
 i) $T(0) = 0$, ii) $T(-x) = -T(x)$ iii) $T(x - y) = T(x) - T(y)$. (07 Marks)

Module-3

- 5 a. Define i) Inner product space ii) Orthogonal sets iii) Orthogonal projections. (06 Marks)
 b. Let $u_1 = \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix}$, $u_2 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ and $y = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ prove that y as the sum of vector in w and a vector orthogonal to w . (07 Marks)

- c. Show that $\{V_1, V_2, V_3\}$ is an orthonormal basis of $\mathbb{R}^3$, where

$$V_1 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix}, \quad V_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad V_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix} \quad (07 \text{ Marks})$$

OR

- 6 a. Use Gram – Schmidt Orthogonalization process to vectors $(1, 0, 1)$, $(1, 0, -1)$ and $(0, 3, 4)$ to obtain an orthonormal basis $\{\alpha_1, \alpha_2, \alpha_3\}$ for $\mathbb{R}^3$. (10 Marks)
 b. Obtain the QR – decomposition for

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}. \quad (10 \text{ Marks})$$

Module-4

- 7 a. Find the Eigen values and Eigen vectors of $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$. (06 Marks)

- b. Obtain the Singular value decomposition of $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}$. (07 Marks)

- c. Diagonalize the following matrix, if possible $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$. (07 Marks)

OR

- 8 a. Explain the quadratic forms. (06 Marks)
 b. Explain the constrained optimization. (07 Marks)
 c. Explain the singular value decomposition. (07 Marks)

Module-5

- 9 a. Define the graphs and networks. (06 Marks)

- b. Draw the multigraph G whose adjacency matrix $M_A = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$. (07 Marks)

- c. Draw the undirected graphs K_3 and K_5 . (07 Marks)

OR

- 10 a. What are the applications of matrices in Engineering? (10 Marks)
 b. Describe the Computer Graphics. (10 Marks)

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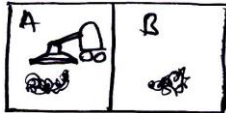
Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Artificial Intelligence

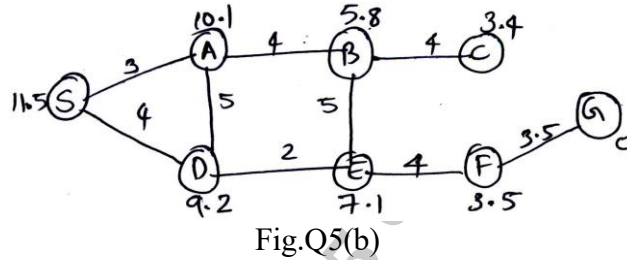
Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the significance of the Turing Test in AI. What abilities does a computer need to pass the turing test? Discuss why AI researchers have not focused extensively on passing the Turing test.	10	L2	CO1
	b.	Write the percept sequence for vacuum cleaner agent and tabulate the workflow of the same with respect to the scenario with location of square A and B as given in Fig.Q1(b).  Fig.Q1(b)	10	L3	CO1
OR					
Q.2	a.	Compare simple reflex agents and model-based reflex agents, focusing on their perception processing, decision-making methods and explain how model-based agents address the limitations of simple reflex agents with their schematic diagrams.	10	L3	CO1
	b.	Analyze and discuss PEAS descriptor for the following applications in detail: i) Medical diagnosis s/m ii) Taxi driver iii) Interactive English tutor iv) Part picking robot v) Refinery controller.	10	L3	CO1
Module – 2					
Q.3	a.	Define Toy problems and Real-world problems in the context of problem-solving approaches with an example for each type in detail.	10	L2	CO2
	b.	Compare and contrast the vacuum world problem and the 8-tile puzzle problems discussing their state representations, initial states, actions and goal tests.	10	L3	CO2
OR					
Q.4	a.	Explain the components and architecture of a problem solving agent.	10	L2	CO2
	b.	Compare and contrast depth-first search with breadth-first search with examples.	10	L3	CO2
Module – 3					
Q.5	a.	Outline a generic knowledge-based agent's program and discuss the difference between declarative and procedural approaches in the context of building knowledge-based agents.	10	L3	CO3

	b.	Apply A* search algorithm to find the solution path from the start node (S) to the goal node (G). The heuristic values (h) are provided with the nodes, and the travel costs (C) are provided with the edges as shown in Fig.Q5(b).	10	L3	CO3
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**OR**

Q.6	a.	Describe the Wumpus world environment and the PEAS specification for the knowledge based agent. Explain how does the agent navigate and make decisions based on percepts in this environment.	10	L2	CO3
	b.	Solve the following eight-tile puzzle using heuristic function approach and the tree diagram considering the initial and final states as specified.	10	L2	CO3

1	2	3
	4	5
7	8	6
Initial State		

1	2	3
4	5	6
7	8	
Final State		

Module – 4

Q.7	a.	Define universal and existential instantiations with examples. Prove the following using Backward and forward chaining : “As per the law, it is a crime for an American to sell weapons to hostile nations. Country E, an enemy of America, has some missiles and all the missiles were sold to it by Solan, who is an American citizen”. Prove that “Solan is a criminal”.	10	L2	CO4
	b.	Explain the following with respect to first-order logic: (i) Assertions and queries (ii) Numbers, sets and lists (iii) The wumpus world.	10	L2	CO4

OR

Q.8	a.	Apply predicate logic to translate and formalize the following statements: (first order logic) (i) Marcus was a man. (ii) Marcus was a Pompeian. (iii) All Pompeian were Romans. (iv) Caesar was a ruler. (v) All Romans were either loyal to Caesar or hated him. (vi) Everyone is loyal to someone. (vii) People only try to assassinate rulers they are not loyal to. (viii) Marcus tried to assassinate Caesar (ix) All men are people. (x) Some people are loyal to Marcus. In each case, provide the appropriate predicates, quantifiers, variables and logical connectives to represent the statements accurately in predicate logic notations.	10	L3	CO4
	b.	Explain backward chaining algorithm with an example.	10	L2	CO4

Module – 5

Module – 5																									
Q.9	a.	In a city, 30% of the population owns a dog, while 70% owns a cat. Among dog owners, 80% take their dogs for daily walks and among cat owners, only 50% do so. If a person is observed walking their pet daily, calculate probability that this person owns a dog. State the Baye’s theorem.	10	L3	CO5																				
	b.	Explain Expert Systems, detailing the characteristics, capabilities, incapacabilities, components and provide two examples.	10	L2	CO5																				
OR																									
Q.10	a.	Explain uncertain knowledge in the context of artificial intelligence. Discuss the challenges an agent focus when acting under uncertainty with the example of diagnosing a dental patient’s toothache.	10	L2	CO5																				
	b.	Explain the concept of inference using full joint probability in the context of agents acting under uncertainty with an example of the following variables: Weather = {sunny, rain, cloudy, snow}, Cavity = {cavity, $\neg$cavity}. Also calculate the following : $P(\text{cavity} \vee \text{toothache})$, $P(\text{cavity} \mid \text{toothache})$, $P(\neg \text{cavity} \mid \text{toothache})$, Given the following full joint distribution for the Toothache, Cavity, Catch world. <table><tr><td></td><td colspan="2">Toothache</td><td colspan="2">$\neg$Toothache</td></tr><tr><td></td><td>Catch</td><td>$\neg$Catch</td><td>Catch</td><td>$\neg$Catch</td></tr><tr><td>Cavity</td><td>0.108</td><td>0.012</td><td>0.072</td><td>0.008</td></tr><tr><td>$\neg$Cavity</td><td>0.016</td><td>0.064</td><td>0.144</td><td>0.576</td></tr></table>		Toothache		$\neg$ Toothache			Catch	$\neg$ Catch	Catch	$\neg$ Catch	Cavity	0.108	0.012	0.072	0.008	$\neg$ Cavity	0.016	0.064	0.144	0.576	10	L3	CO5
	Toothache		$\neg$ Toothache																						
	Catch	$\neg$ Catch	Catch	$\neg$ Catch																					
Cavity	0.108	0.012	0.072	0.008																					
$\neg$ Cavity	0.016	0.064	0.144	0.576																					

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Biology for Engineers (CSE)

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is stem cell? Explain its types and list its applications.	7	L2	CO1
	b.	Explain in detail the properties and functions of nucleic acids.	6	L2	CO1
	c.	Explain the importance of special biomolecules.	7	L2	CO1
OR					
Q.2	a.	What is a biomolecule? Explain the classifications of biomolecule.	7	L2	CO1
	b.	Explain the properties and functions of carbohydrates.	6	L2	CO1
	c.	Describe the structure and functions of a cell with a neat diagram.	7	L3	CO1
Module – 2					
Q.3	a.	What is the role of lipids? Outline the process of obtaining biodiesel from lipids.	7	L3	CO2
	b.	Differentiate between PHA and PLA as a bioplastic materials.	6	L4	CO1
	c.	Explain the role of DNA vaccine for rabies and RNA vaccine for COVID-19.	7	L2	CO1
OR					
Q.4	a.	What are the key properties, advantages and limitations of cellulose based water filters.	7	L3	CO2
	b.	How can DNA finger printing be applied to evaluate its effectiveness and reliability in forensic applications.	6	L4	CO1
	c.	Describe the use of meat analogue and plant protein as food.	7	L2	CO2
Module – 3					
Q.5	a.	Deliberate the functioning of brain as CPU system.	7	L3	CO2
	b.	Write a short note on spirometry and ventilator.	6	L2	CO2
	c.	Explain heart as pump system.	7	L3	CO2
1 of 2					

BBOC407					
OR					
Q.6	a.	Explain eye as a camera system.	7	L3	CO2
	b.	Write a short note on cardiac pacemaker.	6	L2	CO2
	c.	Explain kidney as purification system.	7	L3	CO2
Module – 4					
Q.7	a.	Describe the materials used and engineering applications of Velcro technology.	7	L3	CO3
	b.	Compare the process of photosynthesis to the functioning of photovoltaic cells.	6	L4	CO3
	c.	Explain the HBOCs and PFCs as human blood substituents.	7	L3	CO3
OR					
Q.8	a.	Explain the terms lotus leaf effect and bird flying.	7	L3	CO3
	b.	Compare biological echolocation and technological echolocation highlighting their applications in navigation and detection.	6	L4	CO3
	c.	Explain the terms shark skin, swim suits and bullet train using biological concepts.	7	L3	CO3
Module – 5					
Q.9	a.	Compare the functioning of electrical tongue and human tongue.	7	L4	CO4
	b.	Explain muscle cells as scaffold for tissue growth.	6	L2	CO4
	c.	Explain bioremediation and biomining via microbial surface adsorption.	7	L2	CO4
OR					
Q.10	a.	Illustrate the basic steps of bioprinting process and list the various types of bioprinting techniques.	7	L4	CO4
	b.	Write a short note on: i) Importance of DNA origami ii) Self healing bioconcrete.	6	L2	CO4
	c.	Discuss the applications of artificial intelligence in the diagnosis of disease.	7	L2	CO4

CBCS SCHEME

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BBOK407

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Biology for Engineers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define cell. Explain the structure and function of plant cell with neat diagram.	08	L2	CO1
	b.	Define Stem Cell. Discuss the types and application of stem cells.	06	L2	CO1
	c.	Describe the properties and functions of hormones.	06	L2	CO1
OR					
Q.2	a.	Discuss the properties and functions of nucleic acids in cellular processes.	07	L2	CO1
	b.	Discuss the properties and functions of enzymes.	07	L2	CO1
	c.	Discuss the properties of vitamins and its supplies.	06	L2	CO1
Module – 2					
Q.3	a.	Apply the knowledge of nucleic acid in DNA finger printing in forensic applications.	08	L3	CO1
	b.	Discuss whey protein and plant based protein as protein based food.	06	L2	CO1
	c.	Write a note on PLA as bioplastic.	06	L1	CO1
OR					
Q.4	a.	Apply your knowledge of lipids and outline the process of obtaining biodiesel from lipids.	07	L3	CO1
	b.	Define vaccine. Discuss the mechanism of RNA vaccine for COVID-19.	07	L2	CO1
	c.	Write a note on enzyme based biosensors.	06	L1	CO1
Module – 3					
Q.5	a.	Compare human brain with computer's CPU.	07	L3	CO2
	b.	Explain lungs as a purification system.	07	L2	CO2
	c.	Write a note on dialysis systems of kidney.	06	L1	CO2
OR					
1 of 2					

Q.6	a.	Illustrate the engineering solutions available for Parkinson's disease.	07	L3	CO2
	b.	Explain heart as a pumping system.	07	L2	CO2
	c.	Write a note on optical correction and materials used for lens.	06	L1	CO2
Module – 4					
Q.7	a.	Illustrate the HBOCs and PFCs as human substituents.	07	L3	CO3
	b.	Explain how the structure of shark skin reduces drag and how these properties have been applied to improve swim suit.	07	L2	CO3
	c.	Explain the term GPS and aircrafts technology as bio inspired by bird fly.	06	L2	CO3
OR					
Q.8	a.	Compare the uses of ultrasonography and sonars.	07	L3	CO3
	b.	Discuss the king fisher beak shaped bullet train to the reduction of noise and improve the stability.	06	L2	CO3
	c.	Explain the term superhydrophobic and self-cleaning in lotus leaf effect.	07	L2	CO3
Module – 5					
Q.9	a.	Explain bioimaging and artificial intelligence technique in disease diagnosis.	08	L2	CO4
	b.	Explain the working principles of electrical tongue and electrical nose in food industry.	06	L2	CO4
	c.	Write a note on bioengineering of Muscular dystrophy and osteoporosis.	06	L1	CO4
OR					
Q.10	a.	Explain the process of biomining via microbial surface adsorption.	07	L2	CO4
	b.	Describe the concept of DNA origami and its role in bio-computing.	06	L2	CO4
	c.	Write a note on self healing bio-concrete and bio-mineralization processes.	07	L1	CO4

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Module – 3

Q.5	a.	Explain types capital budgeting decisions.	10	L1	CO2												
	b.	A company is considering investment proposal at a cost of Rs.50000. The facility has a life expectancy of 5 years and no salvage value. If straight line depreciation is followed, compute the following for the data given: i) Pay back period ii) Average rate of return iii) Internal rate of return. <table border="1"> <thead> <tr> <th>Year</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th></tr> </thead> <tbody> <tr> <td>CF BT</td><td>10000</td><td>10,692</td><td>12,769</td><td>13,462</td><td>20,385</td></tr> </tbody> </table>	Year	1	2	3	4	5	CF BT	10000	10,692	12,769	13,462	20,385	10	L2	CO3
Year	1	2	3	4	5												
CF BT	10000	10,692	12,769	13,462	20,385												

OR

Q.6	a.	Compare between conventional and non-conventional cash flows.	10	L3	CO3
	b.	Band box is considering the purchase of a new machine. Two options are available; Low Speed System (LSS) with an initial cost of Rs.20,000 and a High Speed System (HSS) with an initial cost of Rs.30,000. Each system has 15 years of life with no salvage value. CFAT is as follows. Suggest which system should be purchased assume 14% cost of capital. CFAT for years 1 – 15: LSS – Rs.4000/- HSS – Rs.6000/-	10	L2	CO3

Module – 4

Q.7	a.	Explain the factors to be considered while applying Weighted Average Cost of Capital (WACC).	10	L2	CO3
	b.	Amit electronics is evaluating an expansion project that is expected to cost Rs.20 million and generate an annual after tax cash flow of Rs.4 million for the next 10 years. The tax rate is 35%. The company has a target debt-equity ratio of 1:1. Its cost of equity is 16.9%, whereas its pre tax cost of debt is 14%. The floatation cost of equity is 12%, whereas the floatation cost of debt is 2%, calculate the NPV of the expansion project, Solve if it is worth.	10	L2	CO3

OR

Q.8	a.	What factors affect the weighted average cost of capital? Explain.	10	L1	CO2
	b.	What are the common misconceptions surrounding cost of capital in practice? How do you dispel them? Explain.	10	L1	CO3

Module – 5

Q.9	a.	List a few advantages and disadvantages of equity capital.	10	L2	CO3
	b.	Explain features of debentures.	10	L2	CO2

OR

Q.10	a.	How commercial banks contribute towards providing working capital for financing current assets? Explain briefly.	10	L2	CO1
	b.	Compare between permanent and temporary working capital.	10	L2	CO3

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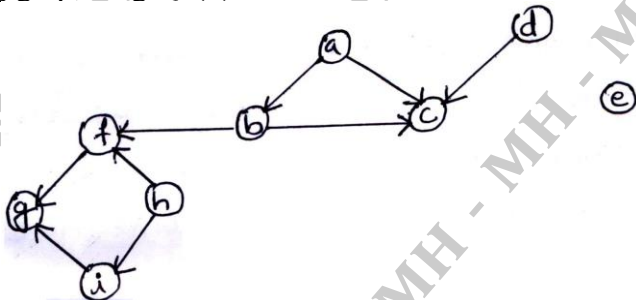
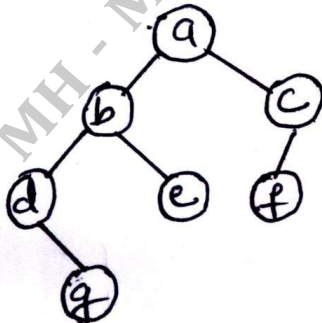
Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Analysis and Design of Algorithms

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the various steps in algorithm design and analysis process with the flow diagram.	08	L1	CO1
	b.	Give formal and informal definitions of asymptotic notations.	06	L1	CO1
	c.	Explain the general plan of mathematical analysis of recursive algorithm with an example.	06	L1	CO1
OR					
Q.2	a.	Design algorithm for tower of Hanoi problem and obtain time complexity.	10	L1	CO1
	b.	Write an algorithm to search an element in an array using sequential search. Discuss the best case, worst case and average case efficiency of this algorithm.	10	L1	CO1
Module – 2					
Q.3	a.	Write an algorithm to sort the numbers using insertion sort. Discuss its efficiency.	10	L2	CO2
	b.	Design quick sort algorithm and obtain its best, average and worst case efficiency.	10	L2	CO2
OR					
Q.4	a.	Write merge sort algorithm and sort the list E X A M P L E.	08	L2	CO2
	b.	Apply the DFS based algorithm to solve the topological sorting problem for the following graph, Fig.Q4(b)	06	L3	CO2
					
Fig.Q4(b)					
	c.	Write algorithm for pre-order, post order and in order traversals of a tree. Write pre-order, in-order and post order for the given tree.	06	L2	CO2
					
Fig.Q4(c)					
1 of 3					

Module – 3

Q.5	a.	Define AVL tree. Construct AVL tree for the list 5, 6, 8, 3, 2, 4, 7.	10	L3	CO3
	b.	Define heap. Sort the following lists by heapsort: H E A P S O R T (in alphabetical order)	10	L3	CO3

OR

Q.6	a.	Write the algorithm for comparison counting sort. Discuss its efficiency.	10	L2	CO4
	b.	Design Horspools algorithm for string matching. Apply Horspools algorithm to find the pattern BARBER on the text JIM_SAW_ME_IN_BARBERSHOP	10	L3	CO4

Module – 4

Q.7	a.	Write Warshall's algorithm and apply the same to compute transitive closure of a directed graph. <table border="1"> <tr> <td></td> <td>a</td> <td>b</td> <td>c</td> <td>d</td> <td>e</td> </tr> <tr> <td>a</td> <td>1</td> <td>0</td> <td>0</td> <td>1</td> <td>0</td> </tr> <tr> <td>b</td> <td>0</td> <td>1</td> <td>0</td> <td>0</td> <td>0</td> </tr> <tr> <td>c</td> <td>0</td> <td>0</td> <td>0</td> <td>1</td> <td>1</td> </tr> <tr> <td>d</td> <td>1</td> <td>0</td> <td>0</td> <td>0</td> <td>0</td> </tr> <tr> <td>e</td> <td>0</td> <td>1</td> <td>0</td> <td>0</td> <td>1</td> </tr> </table>		a	b	c	d	e	a	1	0	0	1	0	b	0	1	0	0	0	c	0	0	0	1	1	d	1	0	0	0	0	e	0	1	0	0	1	10	L3	CO3
	a	b	c	d	e																																				
a	1	0	0	1	0																																				
b	0	1	0	0	0																																				
c	0	0	0	1	1																																				
d	1	0	0	0	0																																				
e	0	1	0	0	1																																				

- b. Construct minimum cost spanning tree using Kruskal's algorithm for the following graph, Fig.Q7(b).

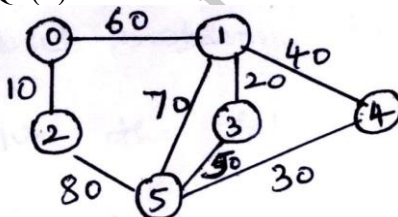


Fig.Q7(b)

OR

Q.8	a.	Solve the following single source shortest path problem assuming vertex '5' as the source. 	10	L3	CO4
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Fig.Q8(a)

- b. Write Huffman's algorithm. Construct Huffman tree and resulting code word for the following:

Character	A	B	C	D	E	-
Probability	0.5	0.35	0.5	0.1	0.4	0.2

Encode the text DAD_CBE.

Module – 5

Q.9	a.	Explain the following with example: (i) P problem (ii) NP problem	06	L1	CO5
	b.	What is decision tree? Construct decision tree for the three element insertion sort.	08	L2	CO5
	c.	Construct state space tree to solve 4 queens problem.	06	L3	CO5

OR

Q.10	a.	What is backtracking? Apply back tracking to solve the below instance of sum of subset problem: $s = \{3, 5, 6, 7\}$, $d = 15$	10	L3	CO6															
	b.	Solve the following instance of knapsack problem using branch and bound technique knapsack capacity = 10. <table><tr><td>Item</td><td>Weight</td><td>Value</td></tr><tr><td>1</td><td>4</td><td>40</td></tr><tr><td>2</td><td>7</td><td>42</td></tr><tr><td>3</td><td>5</td><td>25</td></tr><tr><td>4</td><td>3</td><td>12</td></tr></table>	Item	Weight	Value	1	4	40	2	7	42	3	5	25	4	3	12	10	L4	CO6
Item	Weight	Value																		
1	4	40																		
2	7	42																		
3	5	25																		
4	3	12																		

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Microcontroller

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain the purpose of various fields of current program status register with a neat diagram.	05	L2	CO1
	b.	Explain the ARM design philosophy.	06	L2	CO1
	c.	Explain the core extensions of ARM processor with neat block diagram.	09	L2	CO1
OR					
Q.2	a.	Explain Embedded systems hardware with a neat block diagram.	06	L2	CO1
	b.	What is pipelines in ARM? Illustrate with an example the pipeline stage of ARM 9 and ARM 10.	09	L2	CO1
	c.	Describe the RISC design philosophy with 4 design rules.	05	L2	CO1
Module – 2					
Q.3	a.	Explain the following with examples : (i) RSC (ii) MLA (iii) STRH (iv) SWP	10	L2	CO2
	b.	Explain the different data processing instruction in ARM.	10	L2	CO2
OR					
Q.4	a.	Explain Barrel shifter instruction in ARM with suitable examples.	10	L2	CO2
	b.	Explain the different branch instruction of ARM processor.	05	L2	CO2
	c.	Explain co-processor instruction of ARM processor.	05	L2	CO2
Module – 3					
Q.5	a.	Explain the different basic data types in C. Provide examples of how each data type can be used in a C program.	08	L2	CO3
	b.	Discuss the concept of register allocation in compiler optimization. Illustrate its significance with an example.	07	L2	CO3
	c.	Describe the process of a function call in C.	05	L2	CO3
OR					
Q.6	a.	Discuss the common portability issues faced when writing C programs. How can these issues be mitigated.	07	L2	CO3
	b.	Explain the concept of pointer aliasing with example.	07	L2	CO3
	c.	How are function calls handled efficiently in calling function in C?	06	L2	CO3
Module – 4					
Q.7	a.	What are interrupts? Discuss interrupt vector table with diagram for ARM processor.	06	L2	CO4
	b.	Describe the sequence of operations that occurs when an ARM processor handles an IRQ exceptions.	06	L2	CO4
	c.	Discuss the priority system for exception in ARM processor.	08	L2	CO4
OR					
Q.8	a.	Explain the role of the link register in ARM exception handling.	08	L2	CO4
	b.	Explain the design and implementation of an interrupt stack in a ARM-based system. Explain the steps involved.	08	L2	CO4
	c.	What are the key differences between a boot loader and firmware?	04	L2	CO4

Module – 5					
Q.9	a.	Explain the basic operation of a cache controller.	06	L2	CO5
	b.	With a neat diagram, explain the basic architecture of a cache memory.	10	L2	CO5
	c.	Mention any 4 relationship between cache and main memory.	04	L2	CO5
OR					
Q.10	a.	Write a note on cache write policy both write back or write through.	10	L2	CO5
	b.	Describe the allocation policy on a cache miss.	04	L2	CO5
	c.	Write a note on following : (i) Write buffers (ii) Cache efficiency	06	L2	CO5

* * * * *

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Database Management System

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C																																									
Q.1	a.	Define the following terms: (i) Database (ii) Schema (iii) Entity (iv) DDL (v) Degree of a relationship	05	L1	CO1																																									
	b.	Briefly explain characteristics of database approach.	05	L2	CO1																																									
	c.	List and explain advantages of using DBMS approach.	10	L2	CO1																																									
OR																																														
Q.2	a.	Define the following terms: (i) Cardinality (ii) Weak entity (iii) Program data independence (iv) DML (v) Value sets	05	L1	CO1																																									
	b.	Describe three-schema architecture. Why do we need mappings between schema levels?	05	L2	CO1																																									
	c.	Explain different types of attributes in ER model with suitable example for each.	10	L2	CO1																																									
Module – 2																																														
Q.3	a.	With suitable example, explain the entity integrity and referential integrity constraints. Why each is considered important?	05	L2	CO2																																									
	b.	Discuss equijoin and natural join with suitable example using relational algebra notation.	05	L2	CO2																																									
	c.	Given the relational tables: <table><tr><td colspan="4">Employee:</td><td colspan="2">Department:</td></tr><tr><td>EID</td><td>Name</td><td>DepID</td><td>Salary</td><td>DeptID</td><td>DeptName</td></tr><tr><td>1</td><td>Alice</td><td>10</td><td>5000</td><td>10</td><td>HR</td></tr><tr><td>2</td><td>Bob</td><td>20</td><td>6000</td><td>20</td><td>IT</td></tr><tr><td>3</td><td>Eve</td><td>20</td><td>6500</td><td>30</td><td>Sales</td></tr></table> Project <table><tr><td>PID</td><td>Project Name</td><td>DeptID</td></tr><tr><td>101</td><td>Project Alpha</td><td>10</td></tr><tr><td>102</td><td>Project Beta</td><td>20</td></tr><tr><td>103</td><td>Project Gamma</td><td>30</td></tr></table> Write relational algebra expression for the following: (i) Find the names and salaries of all employees in the ‘IT’ department. (ii) Find the ID’s and names of employees who are in the ‘IT’ department and have a salary greater than 6000. (iii) Find the ID’s and names of employees who are either in the ‘HR’ department or have a salary greater than 6000. (iv) Find the names of employees who are not in the ‘IT’ department (v) Find the names of employees along with their department names.	Employee:				Department:		EID	Name	DepID	Salary	DeptID	DeptName	1	Alice	10	5000	10	HR	2	Bob	20	6000	20	IT	3	Eve	20	6500	30	Sales	PID	Project Name	DeptID	101	Project Alpha	10	102	Project Beta	20	103	Project Gamma	30	10	L3
Employee:				Department:																																										
EID	Name	DepID	Salary	DeptID	DeptName																																									
1	Alice	10	5000	10	HR																																									
2	Bob	20	6000	20	IT																																									
3	Eve	20	6500	30	Sales																																									
PID	Project Name	DeptID																																												
101	Project Alpha	10																																												
102	Project Beta	20																																												
103	Project Gamma	30																																												

1 of 3

OR

Q.4	a.	Explain any two operations that change the state of relation in a database. Provide suitable examples.	05	L2	CO2																																										
	b.	Discuss the aggregation functions and grouping in relational algebra with suitable examples.	05	L2	CO2																																										
	c.	Given the relational tables: <table><tr><th colspan="2">Student:</th></tr><tr><th>SID</th><th>Name</th></tr><tr><td>a</td><td>Alice</td></tr><tr><td>b</td><td>Bob</td></tr><tr><td>c</td><td>Carol</td></tr></table><table><tr><th colspan="2">Project:</th></tr><tr><th>PID</th><th>Project Name</th></tr><tr><td>p</td><td>Alpha</td></tr><tr><td>q</td><td>Beta</td></tr><tr><td>r</td><td>Gamma</td></tr></table><table><tr><th colspan="2">Language:</th></tr><tr><th>LID</th><th>Language Name</th></tr><tr><td>x</td><td>Python</td></tr><tr><td>y</td><td>Java</td></tr><tr><td>z</td><td>C++</td></tr></table><table><tr><th colspan="2">Enrollment:</th></tr><tr><th>SID</th><th>PID</th></tr><tr><td>a</td><td>p</td></tr><tr><td>a</td><td>q</td></tr><tr><td>b</td><td>q</td></tr><tr><td>c</td><td>r</td></tr></table> Write relational algebra expression for the following: (i) Rename the student table to Learner and display it. (ii) Find the students (learners) who are not enrolled in any project. (iii) Find the students who are enrolled in all projects. (iv) Find the students who are not enrolled in any project. (v) Find the students who are enrolled in both the 'Alpha' and 'Beta' projects.	Student:		SID	Name	a	Alice	b	Bob	c	Carol	Project:		PID	Project Name	p	Alpha	q	Beta	r	Gamma	Language:		LID	Language Name	x	Python	y	Java	z	C++	Enrollment:		SID	PID	a	p	a	q	b	q	c	r	10	L3	CO2
Student:																																															
SID	Name																																														
a	Alice																																														
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c	Carol																																														
Project:																																															
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r	Gamma																																														
Language:																																															
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x	Python																																														
y	Java																																														
z	C++																																														
Enrollment:																																															
SID	PID																																														
a	p																																														
a	q																																														
b	q																																														
c	r																																														

Module – 3

Q.5	a.	Explain Armstrong inference rules.	05	L2	CO4
	b.	What is the need for normalization? Explain 1NF, 2NF and 3NF with examples.	05	L2	CO4
	c.	What is functional dependency? Write an algorithm to find minimal cover for set of functional dependencies. Construct minimal cover M for set of functional dependencies which are: $E = \{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$	10	L3	CO4

OR

Q.6	a.	Explain the types of update anomalies in SQL with an example.	05	L2	CO4
	b.	Explain types of JDBC drivers.	05	L2	CO5
	c.	Consider the schema $R = ABCD$, subjected to FDs $F = \{A \rightarrow B, B \rightarrow C\}$, and the non-binary partition $D1 = \{ACD, AB, BC\}$. State whether D1 is a lossless decomposition? [give all steps in detail].	10	L3	CO4

Module – 4

Q.7	a.	Define transaction. Discuss ACID properties.	05	L2	CO5
	b.	With a neat diagram, explain transition diagram of a transaction.	05	L2	CO5
	c.	Demonstrate working of assertion and triggers in SQL with example.	10	L3	CO5

OR

Q.8	a.	Explain cursor and its properties in embedded SQL with suitable example.	05	L2	CO5
	b.	Determine if the following schedule is serializable and explain your reasoning: i) $T1 : R(X)W(X) \quad T2 : R(X)W(X) \quad T1 : COMMIT \quad T2 : COMMIT$ ii) $T1 : W(X)R(Y) \quad T2 : R(X)W(Y) \quad T1 : COMMIT \quad T2 : COMMIT$	05	L2	CO5

	c.	Consider the tables below: Sailors (<u>sid</u> : integer, sname : string, rating : integer, age : real) Boats (<u>bid</u> : integer, bname : string, color : string); Reserves (<u>sid</u> : integer, <u>bid</u> : integer, day : date) Write SQL queries for the following: (i) Write create table statement for reserves. (ii) Find all information of sailors who have reserved boat number 101. (iii) Find the names of sailors who have reserved at least one boat. (iv) Find the names of sailors who have reserved a red boat. (v) Find the average age of sailors for each rating level.	10	L3	CO5
Module – 5					
Q.9	a.	Explain the CAP theorem.	05	L2	CO6
	b.	What is NOSQL graph database? Explain Neo4j.	05	L2	CO6
	c.	Why concurrency control and recovery are needed in DBMS? Demonstrate with suitable examples types of problems that may occur when two simple transactions run concurrently.	10	L3	CO5
OR					
Q.10	a.	Explain basic operations CRUD in MongoDB.	05	L2	CO6
	b.	Explain deadlock prevention protocols.	05	L2	CO5
	c.	Briefly discuss the two-phase locking techniques for concurrency control.	10	L3	CO5

* * * * *

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Discrete Mathematical Structures

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define a tautology. Prove that for any propositions p, q, r the compound propositions $\{(p \rightarrow q) \wedge (q \rightarrow r)\} \rightarrow (p \rightarrow r)$ is tautology.	06	L2	CO1
	b.	Establish the validity of the following argument using the rules of inference: $\{p \wedge (p \rightarrow q) \wedge (s \vee r) \wedge (r \rightarrow \sim q)\} \rightarrow (s \vee t)$	07	L2	CO1
	c.	For any two odd integers m and n, show that: (i) $m + n$ is even (ii) mn is odd	07	L2	CO1
OR					
Q.2	a.	Show that the compound proposition $[(p \vee q) \rightarrow r] \Leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)]$ for primitive statements p, q, r is logically equivalent.	06	L2	CO1
	b.	Prove the following using law of logic: $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$	07	L2	CO1
	c.	Determine the truth value of each of the following quantified statements, the universe being the set of all non-zero integers: (i) $\exists x, \exists y, [xy = 1]$ (ii) $\exists x, \forall y, [xy = 1]$ (iii) $\forall x, \exists y, [xy = 1]$ (iv) $\exists x, \exists y, [(2x + y = 5) \wedge (x - 3y = -8)]$ (v) $\exists x, \exists y, [(3x - y = 17) \wedge (2x + 4y = 3)]$	07	L3	CO1
Module – 2					
Q.3	a.	Prove that for each $n \in \mathbb{Z}^+$, $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.	06	L2	CO2
	b.	Let $a_0 = 1, a_1 = 2, a_2 = 3$ and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$, prove that $a_n \leq 3^n \forall n \in \mathbb{Z}^+$.	07	L2	CO2
	c.	How many positive integers n can be we form using the digits 3, 4, 4, 5, 5, 6, 7 if we want n to exceed 5,000,000?	07	L3	CO2
OR					
Q.4	a.	By mathematical induction prove that $1.3 + 2.4 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$	06	L2	CO2
	b.	Find the number of permutations of the letters of the word ENGINEERING such that: (i) All the E's are together (ii) Arrangement begin with N (iii) All the vowels are adjacent.	07	L3	CO2
	c.	Find the coefficient of $a^2b^3c^2d^5$ in the expansion of $(a + 2b - 3c + 2d + 5)^{16}$.	07	L3	CO2
Module – 3					
Q.5	a.	State pigeon hole principle. Prove that if 30 dictionaries in a library contain a total of 61,327 pages then atleast one of the dictionaries must have atleast 2045 pages.	06	L3	CO3
	b.	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 3x-5 & \text{if } x > 0 \\ 1-3x & \text{if } x \leq 0 \end{cases}$. Find $f^{-1}(0), f^{-1}(1), f^{-1}(-1), f^{-1}(3), f^{-1}(-6), f^{-1}([-6, 5])$ and $f^{-1}([-5, 5])$	07	L2	CO3
	c.	Draw the Hasse diagram representing the positive divisor of 36.	07	L3	CO3

OR

Q.6	a.	Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6\}$, (i) How many functions are there from A to B? (ii) How many of these are one to one? (iii) How many functions are there from B to A? (iv) How many of these are onto?	06	L2	CO3
	b.	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$. If $(g \circ f)(x) = 9x^2 - 9x + 3$, determine a and b .	07	L2	CO3
	c.	Let $A = \{1, 2, 3, 4, 6\}$ and R be a relation on A defined by aRb if and only if “ a is multiple of b ”. Write down the relation R , relation matrix $M(R)$ and draw the digraph. List out in degree and out degree.	07	L3	CO3

Module – 4

Q.7	a.	In how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block?	06	L3	CO4
	b.	Determine the number of positive integers n such that $1 \leq n \leq 100$ and n is not divisible by 2, 3, or 5.	07	L3	CO4
	c.	Solve the recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0$, $a_0 = 1$, $a_1 = 6$.	07	L2	CO4

OR

Q.8	a.	In how many ways can the 26 letters of the English alphabet be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?	06	L3	CO4
	b.	Five teachers T_1, T_2, T_3, T_4 are to be made class teachers for five classes, C_1, C_2, C_3, C_4, C_5 , one teacher for each class. T_1 and T_2 do not wish to become the class teachers for C_1 or C_2 , T_3 and T_4 for C_4 or C_5 , and T_5 for C_3 or C_4 or C_5 . In how many ways can the teachers be assigned the work? (Without displeasing any teacher)	07	L3	CO4
	c.	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$ where $n \geq 0$ and $F_0 = 0$, $F_1 = 1$.	07	L2	CO4

Module – 5

Q.9	a.	If G be a set of all non zero real numbers and let $a * b = \frac{ab}{2}$ then show that $(G, *)$ is an abelian group.	06	L2	CO5
	b.	Define Klein group and if $A = \{e, a, b, c\}$ then show that this is a Klein-4 group.	07	L2	CO5
	c.	State and prove Lagrange's theorem.	07	L2	CO5

OR

Q.10	a.	If H and K are subgroups of group G, prove that $H \cap K$ is also a subgroup of G. Is $H \cup K$ a subgroup of G?	06	L2	CO5																																																	
	b.	Define cyclic group and show that $(G, *)$ whose multiplication table is as given below is cyclic. <table border="1"><tr><td>*</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>a</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>b</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td><td>a</td></tr><tr><td>c</td><td>c</td><td>d</td><td>e</td><td>f</td><td>a</td><td>b</td></tr><tr><td>d</td><td>d</td><td>e</td><td>f</td><td>a</td><td>b</td><td>c</td></tr><tr><td>e</td><td>e</td><td>f</td><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>f</td><td>f</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr></table>	*	a	b	c	d	e	f	a	a	b	c	d	e	f	b	b	c	d	e	f	a	c	c	d	e	f	a	b	d	d	e	f	a	b	c	e	e	f	a	b	c	d	f	f	a	b	c	d	e	07	L2	CO5
*	a	b	c	d	e	f																																																
a	a	b	c	d	e	f																																																
b	b	c	d	e	f	a																																																
c	c	d	e	f	a	b																																																
d	d	e	f	a	b	c																																																
e	e	f	a	b	c	d																																																
f	f	a	b	c	d	e																																																
	c.	Let $G = S_4$, for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$, find the subgroup $H = \langle \alpha \rangle$. Determine the left cosets of H in G.	07	L3	CO5																																																	

CBCS SCHEME

BCS456B

USN

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Capacity Planning for IT

Time: 1 hrs.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
 2. Use only **Black ball point pen** for writing / darkening the circles.
 3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
 4. Darkening two circles for the same question makes the answer invalid.
 5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.
-
1. Which metric is NOT mentioned as part of defining an application's load and capacity requirements?
 - a) Response times
 - b) Consumable capacity
 - c) Peak driven processing
 - d) User satisfaction scores
 2. According to the textbook, what is an example of peak driven processing?
 - a) Increased mobile traffic
 - b) Daily routine operations of a web application
 - c) Spike in tweets during super bowl
 - d) Continuous monitoring of system performance
 3. What is one of the challenges of capacity planning in data centers mentioned in the text?
 - a) Short supply-chain cycles
 - b) High cost of elasticity
 - c) Long supply-chain cycles and configuration overload
 - d) Ease of horizontal scaling
 4. Identify what does "graceful degradation" refer to
 - a) Complete system shutdown
 - b) Slightly reduced performance under increased load
 - c) Sudden failure without warning
 - d) Manual intervention to prevent failure
 5. Interpret why is it important to tie server statistics to actual site or mobile app metrics?
 - a) To reduce the need for hardware upgrades
 - b) To better understand and track how capacity will change over time
 - c) To ensure data privacy and security
 - d) To simplify the architecture of the system

6. Recognize which approach is suggested to manage unexpected spikes in traffic without always running maximum capacity?
 - a) Reducing the number of active servers
 - b) Implementing queuing and caching based approaches
 - c) Decreasing the QoS (quality of service)
 - d) Limiting the number of users accessing the service
7. Why is collaboration with product and engineering teams crucial in capacity planning?
 - a) To reduce the overall workload
 - b) To align capacity requirements with long term roadmaps and predict need accurately
 - c) To minimize the need of new hardware
 - d) To ensure the security of the system
8. What is a potential risk of not having adequate capacity ready during traffic spikes?
 - a) Increased operational costs
 - b) Improved user experience
 - c) Violations of SLA's and adverse impacts on user experience
 - d) Reduced data redundancy
9. Recognize the purpose of using graphs and measurement tools in capacity planning.
 - a) To simplify the architecture of the system
 - b) To visually represent system performance and assist in decision making
 - c) To reduce the need for technical expertise
 - d) To limit the amount of data collected
10. In the context of cloud infrastructure, what does "Cloud Jail" refer to?
 - a) A security issue where data is trapped and inaccessible
 - b) The scenario where a company is excessively spending on cloud services and is dependent on the cloud provider
 - c) A situation where cloud services are unavailable due to downtime
 - d) The restriction of cloud services to a single geographic location
11. Which of the following used in distributed monitoring software?
 - a) Gomez
 - b) Ganglia
 - c) Elastr
 - d) None of these
12. Identify the primary purpose of capacity tracking tools.
 - a) To generate alerts concerning system availability
 - b) To automatically and routinely measure server behaviour over a predefined amount of time
 - c) To manage network configuration
 - d) To install and update software on servers
13. Which of the following tools is NOT typically used for load testing?
 - a) Load runner
 - b) Iago
 - c) JMeter
 - d) Grafana
14. Recognise the tool which is significantly designed for querying and usualizing time-series operational metrics?
 - a) Ganglia
 - b) Grafana
 - c) Graphite
 - d) Cacti
15. What protocol is commonly used for gathering metrics for most networking and server equipment?
 - a) FTP
 - b) TCP/IP
 - c) SNMP
 - d) HTTP

16. Which of the following is NOT an example of an application level metric mentioned in the text?
- a) Number of tweets per minute
 - b) Number of messages per minute
 - c) CPU usage per server
 - d) Number of photos uploaded per minute
17. Recognize the key metric that should be tracked to predict when more storage hardware needs to be ordered?
- a) Total disk space consumption over time
 - b) Number of file accesses per minute
 - c) Disk I/O operations per second
 - d) CPU utilization on storage servers
18. In the real-world example provided, which days were identified as peak periods for photo uploads in Flickr?
- a) Fridays
 - b) Saturdays
 - c) Sundays
 - d) Mondays
19. What defines the capacity ceiling for a cache with a small or slowly increasing working set?
- a) Network usage
 - b) Hit ratio
 - c) CPU usage
 - d) Disk I/O utilization
20. Identify the resource system metric which is the defining the peak resource metric for a SQUID caching server.
- a) Memory usage
 - b) Network usage
 - c) Disk I/O wait
 - d) CPU usage
21. Select among which of the following requirements and capacity goals are interrelated with performance, availability and reliability
- a) External service monitoring
 - b) Business requirements
 - c) User expectations
 - d) All of these
22. Among the following metrics recognize the ones used by commercial services to monitor site performance?
- a) Page load time
 - b) Time to first byte (TTFB)
 - c) Secure Sockets Layer (SSL) time
 - d) All of these
23. Interpret which among the following monitoring services are considered at enterprise level
- a) Site uptime
 - b) Alletra
 - c) Ping Dom
 - d) Keynote
24. Use the following data and answer, "If a site generates \$ 3000 worth of sales every minute, how much revenue could potentially be lost during a 10 minute downtime, assuming no customer recovery".
- a) \$ 30000
 - b) \$ 15000
 - c) \$ 45000
 - d) \$ 3000
25. Identify the root cause of perceived slowness of a web page easier in monolithic architecture than in Service-Oriented Architecture (SOA) or Microservice (MSA) Architecture
- a) Monolithic architectures are more modern
 - b) Monolithic architectures use more third-party vendors
 - c) SOA and MSA do not use high quality images
 - d) SOA and MSA have complex interactions between different microservices.

26. What is a common usage of the term “five-nines” in SLA discussions?
- a) To describe a feature set
 - b) To indicate high availability of a site or system
 - c) To refer to the numbers of servers
 - d) To measure bandwidth capacity
27. What is vertical scaling? Discuss
- a) Adding more servers to share the load
 - b) Increasing the resources within a single server
 - c) Using cloud services for additional capacity
 - d) Reducing the number of servers in use
28. What was the primary benefit of diagonal scaling in the Yahoo! company? Interpret.
- a) Decreased disk usage
 - b) Increased network bandwidth
 - c) Reduced load average and power consumption
 - d) Improved security
29. What does the term “SPOF” stand for?
- a) Single point of failure
 - b) Scalable point of function
 - c) Standard protocol of functionality
 - d) System performance optimization framework
30. Interpret the process of curve fitting mean
- a) Finding a mathematical equation that mimics the data
 - b) Reducing the number of servers
 - c) Enhancing network speed
 - d) Increasing CPU performance
31. Realize the meaning of procurement pipeline
- a) The process by which we obtain new capacity
 - b) The process of developing new software
 - c) The process of hiring new employees
 - d) The process of marketing new product
32. Discuss in which step of the procurement process it involves creating of graphs with attention grabbing fonts?
- a) Testing
 - b) Justify purchases
 - c) Determine the needs
 - d) Solicit quotes from vendors
33. What does an Amazon Machine Image (AMI) include?
- a) Only the OS
 - b) Only application server
 - c) A template for the root volume, launch permissions, block device mapping
 - d) A template of the root volume and application servers only
34. In the context of configuration management what is a “role”? Discuss.
- a) A particular type of machine that performs a specific task
 - b) A collection of unrelated software and configuration
 - c) A single server dedicated to a unique task
 - d) A temporary setting used for testing purpose

35. What is one advantage of image based systems for OS installations? Identify from following.
- a) Greater accountability for installed files
 - b) Faster installation process
 - c) Uses package management tools
 - d) More complex to manage
36. What is the “Holy grail” of all deployment systems according to the text?
- a) Have administrators manually install software on new server
 - b) Have new servers automatically start working without administrator intervention
 - c) Keep hardware inventory in a distributed manner
 - d) Perform all configuration changes directly on individual servers
37. Realize why should logging into an individual server be restricted?
- a) To prevent unauthorized access
 - b) To make it harder to troubleshoot
 - c) To allow different configuration settings for different hardware
 - d) To ensure all management is done from a central location
38. What should be done if hardware limitations change?
- a) Continue using the old ceilings
 - b) Update the metric used to determine the ceiling
 - c) Ignore the changes until the next review
 - d) Focus only on software improvement
39. Understanding user behaviour is important in capacity planning. Justify.
- a) It provides valuable data for predicting future demand.
 - b) It ensures better marketing strategies
 - c) It reduces the need for technical support
 - d) It helps determine company’s financial health
40. Which metric would you track to understand user engagement in an online community?
- a) Network speed
 - b) CPU usage
 - c) Posts per day, per user
 - d) Network speed
41. What is key benefit of migrating guest OS instances in virtualization?
- a) Improved network performance
 - b) Enhanced application compatibility
 - c) Fault tolerance and hardware failure resilience
 - d) Reduced power consumption
42. What is potential disadvantage of virtualization in terms of capacity planning? Identify.
- a) Complicated metric collection
 - b) Inability to migrate VMs
 - c) Reduced network performance
 - d) Increased power consumption
43. How do containers differ from VMs in terms of resource consumption?
- a) Containers require more memory than VMs
 - b) VMs share the host OS Kernel, while containers do not
 - c) VMs are more portable cross different host environment
 - d) Containers run processes directly on the host OS without a hypervisor

44. What distinguishes containers from VMs in term of application probability?
- a) VMs are more portable across different hypervisors
 - b) Containers provide stronger isolation between application
 - c) VMs share the host OS Kernel of running applications
 - d) Containers are less dependent on underlying hardware
45. In cloud computing, the client is on which end?
- a) Backend
 - b) Front end
 - c) Depends
 - d) Both a and b
46. On what is cloud computing based
- a) Modulation
 - b) Hybridization
 - c) Visualization
 - d) Multiplexing
47. Which of the following is NOT among the building blocks of the best purchase every successful cloud computing. Platform should follow?
- a) World class security
 - b) Complete disaster recovery
 - c) High availability
 - d) Resource utilization
48. Identify which is best known service model.
- a) SaaS
 - b) PaaS
 - c) IaaS
 - d) All of these
49. When you add a software stack, such as an operating system and application to the service the model shifts to _____ model.
- a) SaaS
 - b) PaaS
 - c) IaaS
 - d) All of these
50. Which of the following is cloud platform of Amazon? Identify.
- a) Azure
 - b) AWS
 - c) Cloudera
 - d) All of these

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
UI/UX

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Identify which of the following statements best describes the relationship between usability and user experience (UX)?
 - a) Usability is a subset of UX, focusing primarily on how easy and efficient it is for users to achieve their goals.
 - b) UX is a subset of usability, focusing primarily on the technical performance of a product.
 - c) Usability and UX are unrelated concepts, with usability concerning functionality and UX concerning aesthetics only.
 - d) Usability solely determines the emotional impact a product has on its users, which is the essence of UX.
 2. Identify how does emotional impact play a role in user experience (UX)?
 - a) Emotional impact is irrelevant to UX as it only concerns the technical aspects of a product.
 - b) Emotional impact is the primary factor in UX, disregarding usability and functionality.
 - c) Emotional impact complements usability and functionality by enhancing user satisfaction and engagement.
 - d) Emotional impact solely defines the usability of a product.

3. Identify why is it important to build a business case for user experience (UX)?
 - a) A business case for UX is unnecessary because UX improvements do not translate into financial gains.
 - b) Building a business case for UX helps in securing funding and support from stakeholders by demonstrating potential returns on investment through improved customer satisfaction and loyalty.
 - c) A business case for UX should focus solely on the aesthetic enhancements of a product.
 - d) The business case for UX should only be considered after the product has failed in the market.
4. Identify which of the following best describes the role of emotional impact in user experience (UX)?
 - a) Emotional impact is the primary determinant of a products usability.
 - b) Emotional impact enhances the overall user experience by creating positive feelings and engagement with the product.
 - c) Emotional impact is only relevant during the initial interaction with a product.
 - d) Emotional impact is less important than technical performance and functionality in UX.
5. Identify how can emotional impact influence user behavior and perception in the context of user experience (UX)?
 - a) Emotional impact has no influence on user behavior and perception.
 - b) Positive emotional impact can increase user satisfaction, loyalty and the likelihood of recommending the product to others.
 - c) Emotional impact only affects the aesthetic appreciation of a product, not user behavior.
 - d) Negative emotional impact always leads to increased usability issues.
6. Identify what is a key consideration when aiming to create a positive emotional impact in user experience design?
 - a) Focus solely on the visual design elements.
 - b) Understand the target audience's needs, preferences and emotional triggers to design experiences that resonate with them.
 - c) Prioritize functionality over any emotional considerations.
 - d) Avoid any form of user feedback during the design process.
7. Identify what is the first step in extracting interaction design requirements for a UI/UX project?
 - a) Developing high-fidelity prototypes.
 - b) Conducting user research to understand the target audience's needs, behaviors and pain points.
 - c) Creating a detailed project timeline.
 - d) Writing code for the initial version of the product.
8. Identify which method is commonly used to gather user needs and requirements for interaction design?
 - a) A/B testing
 - b) Usability testing
 - c) User interviews and surveys
 - d) Load testing

9. Identify why is it important to involve stakeholders in the process of extracting interaction design requirements?
- a) To ensure that the design aligns with business goals and user needs.
 - b) To decrease the amount of user research needed.
 - c) To speed up the development process by reducing feedback loops.
 - d) To limit the number of design changes during development.
10. Identify what role does creating persona's play in the process of defining interaction design requirements?
- a) Personas are used to test the technical performance of a product.
 - b) Personas help designers understand and visualize the target users, characteristics, needs and goals, guiding design decisions.
 - c) Personas are primarily used to track project timelines and deadlines.
 - d) Persona's are used to develop coding standards for the development team.
11. Design concept includes _____
- a) Usability
 - b) Accessibility
 - c) Both (a) and (b)
 - d) None of these
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 - c) Personas are primarily used to track project timelines and deadlines.
 - d) Personas are used to develop coding standards for the development team.
15. Identify how do usability tests contribute to refining interaction design requirements?
- a) They eliminate the need for user feedback in the design process.
 - b) They provide direct insights into how real users interact with the design, highlighting areas that need improvement.
 - c) They focus exclusively on the visual aspects of the design.
 - d) They are used only after the final product has been developed.

16. Identify which of the following is a key outcome of properly defined interaction design requirements?
 - a) A visually appealing user interface with minimal functionality.
 - b) An efficient development process with no need for revisions.
 - c) A user-centered design that effectively meets user needs and enhances the overall user experience.
 - d) A decrease in overall project costs.
17. Identify which of the following is a common method used for formal requirement extraction in UI/UX?
 - a) Random sampling
 - b) Stakeholder interviews and workshops.
 - c) Creating high-fidelity prototypes before gathering any user input.
 - d) Implementing the first idea that comes to mind.
18. Which of the following tools is often used to gather user requirements in a structured manner?
 - a) Wireframes
 - b) User personas
 - c) Requirements management software like JIRA or Trello
 - d) CSS frameworks like Bootstrap.
19. Identify what happens if the user has a different mental model than the designer?
 - a) The user will have no impact on the systems usability.
 - b) The designer must redesign the system immediately.
 - c) The system will automatically adapt to the user's mental model.
 - d) The user may misunderstand how to use the system effectively.
20. Identify what is a designer's mental model?
 - a) Physical model of the system's hardware.
 - b) Conceptualization of the envisioned system by the designer.
 - c) List of tasks the system should perform.
 - d) User's understanding of how the system works.
21. Identify what is the main role of a visual designer during the visual design and visual comps phase?
 - a) To create wireframe sequences representing key workflows.
 - b) To annotate wireframes with call-out text and data elements.
 - c) To write detailed documentation language for the design.
 - d) To produce pixel-perfect mockups of the graphical look and feel of the user interface.
22. Identify in the context of intermediate design, what are information objects primarily meant to do?
 - a) Store user interface layouts and graphics.
 - b) Provide a visual representation of system data.
 - c) Be accessed and manipulated by users in their work roles.
 - d) Define the system's color scheme.

23. Identify during intermediate design iteration, high-fidelity mockups can serve to:
- Test the final design before launch.
 - Create visual compositions for marketing purposes.
 - Enhance the systems performance.
 - Demonstrate leading conceptual design candidates to upper management.
24. Identify what is a key advantage of using illustrated scenarios in intermediate interaction design documents?
- They are suitable only for initial design stages.
 - They eliminate the need for textual descriptions.
 - They focus only on visual elements without textual support.
 - They combine visual and textual communication to share details.
25. Identify what is meant by 'visual comps' in the context of visual design?
- Early wireframes defining user workflows.
 - Conceptual sketches without precise details.
 - Pixel-perfect mockups of the graphical interface.
 - Annotated wireframes for iteration and evaluation.
26. Identify what are wireframes primarily used for in the field of interaction design?
- Creating high-fidelity visual representations.
 - Documenting, communicating and prototyping interaction designs.
 - Developing complex algorithms for software.
 - Implementing final website designs.
27. Identify which type of prototype is commonly used during the detailed design iteration?
- Low-fidelity wireframes
 - Paper prototypes
 - High-fidelity interactive mockups
 - Conceptual sketches
28. Identify what does the term 'information object' refer to, based on the context provided?
- A wireframe element used for screen layouts.
 - A visual representation of design concepts.
 - A navigation element for back and forth movements.
 - An internally stored work object shared by users and the system.
29. Identify what is one of the most important aspects to remember about wireframing?
- Detailing
 - Modularity
 - Complexity
 - Consistency
30. Identify which user experience goal example is specifically mentioned for a life-critical system?
- Ease-of-use
 - Safety
 - Walk-up-and-use learnability
 - Avoiding errors for intermittent users.
31. Identify what is the main purpose of a horizontal prototype?
- To provide in-depth detail of a specific feature.
 - To optimize the front-end and back-end integration.
 - To test the physical interaction of the design.
 - To provide an overview of many features with limited functionality.

32. Identify which of the following best describes a 'T' prototype?
- a) It offers a shallow overview with detailed development in a few areas.
 - b) It represents the full system but in a low-fidelity format.
 - c) It focuses exclusively on the depth of a single feature.
 - d) It includes a 3D physical mockup for tangible interaction.
33. Identify why might a team use a local prototype?
- a) To evaluate specific isolated design details.
 - b) To simulate the entire system workflow in high fidelity.
 - c) To provide a fully functional back-end system.
 - d) To combine physical and software elements of a design.
34. Identify what is a significant benefit of using physical prototypes over graphical representations on a screen?
- a) Users can access more detailed information.
 - b) Users experience less usage interference.
 - c) Users find it faster to visualize ideas.
 - d) Users can hold and feel the product as a real object.
35. Identify why are low-fidelity prototypes often kept private within the project team?
- a) They require too much time to update frequently.
 - b) They do not provide any useful feedback.
 - c) They can be misinterpreted by users and customers if not carefully explained.
 - d) They are too expensive to produce for public use.
36. Identify what is the main difference between horizontal and vertical prototypes?
- a) Vertical prototypes offer a shallow design overview, while horizontal prototypes provide in-depth feature details.
 - b) Horizontal prototypes cover a broad range of features with less depth, while vertical prototypes focus extensively on a narrow range of features.
 - c) Horizontal prototypes are fully functional, while vertical prototypes are not.
 - d) Horizontal prototypes are used for localized design issues, while vertical prototypes include full system functionality.
37. Identify what is the primary benefit of using a T prototype?
- a) It provides the lowest possible fidelity for initial designs.
 - b) It entirely replaces the need for horizontal and vertical prototypes.
 - c) It combines both horizontal and vertical prototyping advantages by providing a broad overview with detailed functionality in some areas.
 - d) It is only useful for isolated design issues like a local prototype.
38. Identify why might a design team choose to use a low-fidelity prototype?
- a) Low-fidelity prototypes look exactly like the final system, providing realistic evaluations.
 - b) Low-fidelity prototypes are used exclusively when producing advance sales demos.
 - c) Low-fidelity prototypes allow quick construction and iteration, making them cost-effective and fast to produce.
 - d) Low-fidelity prototypes are required whenever final implementation is imminent.

39. Identify which advantage is described for placing paper prototypes inside mobile devices?
- a) It eliminates the need for any user input.
 - b) Users can interact with the device physically.
 - c) The device will function without any power source.
 - d) It guarantees the prototype is free from any errors.
40. Identify why low-fidelity prototypes should be explained carefully to non-team members?
- a) They require special equipment to understand.
 - b) They are too advanced for the early stages of design.
 - c) They contain highly sensitive information.
 - d) They can be misinterpreted by those not familiar with their use.
41. Identify what are the UX design guidelines used for?
- a) Interpreting design guidelines
 - b) Planning
 - c) Translation
 - d) All of these
42. Identify what is a crucial factor to consider when designing a UX environment?
- a) The color of the interface
 - b) The length of the content
 - c) Human memory limitations
 - d) Who the intended audience is
43. Identify in the context of UX design, what does the term 'Physical Action' refer to?
- a) Incorporating interactive elements into the design
 - b) How the user interacts with the interface
 - c) The orientation of the device
 - d) The physical action the user needs to take to use the product.
44. Identify what comes under the assessment in UX Design Guidelines?
- a) Assessing the design guidelines
 - b) Assessing the outcomes
 - c) Assessing the plan
 - d) All of these
45. Identify what does the term 'Translation' refer to in UX Design?
- a) Translating the design into different languages.
 - b) Transferring the inputs to outputs.
 - c) The conversion of users actions and requirements into design elements.
 - d) Altering the design based on user location.
46. Identify what is the function of 'Outcomes' in UX design?
- a) To plan the next steps
 - b) To measure the success of the design
 - c) To analyze user interaction
 - d) To interpret design guidelines
47. Identify what do UX design guidelines help with?
- a) Improve usability and accessibility
 - b) Increase interface complexity
 - c) Limit design creativity
 - d) Increase the time users spend on the site
48. Identify an example of translation in UX design could be?
- a) Storyboarding
 - b) User flow diagrams
 - c) Wireframing
 - d) Prototyping

49. Identify what does planning in UX design involved?
- a) Brainstorming sessions
 - b) Scoping the project
 - c) Deciding the user interface design
 - d) All of these
50. Identify why are human memory limitations considered while designing a UX environment?
- a) To plan the spacing on the page
 - b) To limit the use of multimedia
 - c) To design the interface in a way that doesn't overload the users memory capacity.
 - d) To decide the colors used in the design.

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Advance Java

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1				M	L	C
Q.1	a.	What is collection framework? Explain the methods define by the collection interface.		7	L2	CO1
	b.	Implement a java program to demonstrate creating an ArrayList, adding elements, removing elements, sorting elements of ArrayList.		8	L4	CO1
	c.	Demonstrate with an example how accessing a collection Via an Iterator can be done.		5	L2	CO1
OR						
Q.2	a.	Explain the following map classes : (i) HashMap (ii) TreeMap.		8	L2	CO1
	b.	Explain ArrayList class and explain the following methods : (i) binarysearch (ii) Copy (iii) equals (iv) fill		8	L2	CO1
	c.	Explain legacy classes.		4	L2	CO1
Module – 2						
Q.3	a.	What is string in Java? Implement a Java program to illustrate the use of any four constructors of string class.		8	L3	CO2
	b.	Explain the following stringBuffer methods with an example : (i) insert (ii) append (iii) replace (iv) substring.		8	L2	CO2
	c.	Differentiate between equals () and == with respect to string comparison.		4	L2	CO2
OR						
Q.4	a.	Explain the following string buffer class with examples : (i) Capacity (ii) Reverse (iii) CharAt () (iv) deleteCharAt ().		8	L2	CO2
	b.	Implement a java program to remove duplicate characters from a given string and display the resultant string.		8	L3	CO2
	c.	How CompareTo () method differs from CompareToIgnoreCase () method?		4	L2	CO2
Module – 3						
Q.5	a.	What are differences between swing and AWT? Why Swing components are called light weight component?		8	L2	CO3
	b.	Explain the following with suitable code : JButton, JLabel, JTextField, JComboBox.		12	L2	CO3
OR						
Q.6	a.	Explain the paint package in swings.		6	L2	CO3

	b.	Create a swing application having two buttons named alpha and beta. When either of buttons pressed, it should display “alpha pressed” and “beta pressed” respectively.	10	L4	CO3
	c.	Write short note on MVC connection.	4	L2	CO3
Module – 4					
Q.7	a.	Explain the life cycle of Servlet.	4	L2	CO4
	b.	Describe the core interfaces that are provided in Javax, Servlet, http package.	8	L2	CO4
	c.	Implement a Servlet program to display the name, USN and total marks by accepting student detail.	8	L3	CO4
OR					
Q.8	a.	What is JSP? Explain the various types of JSPtags with examples.	10	L2	CO4
	b.	What are cookies? How cookies are handled in JSP? Write a JSP program to create and read cookie.	10	L3	CO4
Module – 5					
Q.9	a.	What are database drives? Explain the different JDBC driver types.	6	L2	CO5
	b.	Describe the various steps of JDBC with code snippets.	10	L2	CO5
	c.	Write any two syntax of established a connection to a database.	4	L2	CO5
OR					
Q.10	a.	Describe the following concepts : (i) Scrollable Resultset (ii) Callable statement. (iii) Transaction processing (iv) Updatable Resultset.	10	L2	CO5
	b.	What is connection pooling? Explain connection pooling with code snippets.	6	L2	CO5
	c.	Explain different kinds of exceptions in Database.	4	L2	CO5

GBCS SCHEME

BUHK408

USN

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

1. What is the state of liking and a holistic and all-encompassing state of the mind that crates inner harmony?
a) Prosperity b) Happiness c) Innateness d) Organized
2. _____ is defined as a group of people working together to create a surplus
a) Alone b) Organisation c) System d) Army
3. The process whereby national economies and business systems are becoming deeply interlinked with each other is called
a) Nationalization b) Glocalization c) Global linking d) Globalization
4. Which of the following actions will likely lead to organizational ethical behavior?
a) Establishing an ethics office
b) Promoting moral courage
c) Developing strong governance processes
d) All of these
5. _____ behavior tends to arise when managers decide to put the attainment of their own personal goals, or the goals of the organization, above the fundamental rights of one or more stakeholder groups
a) Complementary b) Situational c) Unethical d) Confusing
6. Accepted principles of right or wrong governing the conduct of business people are called :
Choose one answer
a) Business b) Conduct c) Business ethics d) Principles
7. What is true about a professional
a) The professional has to have autonomy
b) He cannot be supervised by the client
c) He should not subject to political control
d) All of these

8. The basic rule of professionalism is “not knowingly to do _____”
a) Harm b) Work c) Competition d) Copying
9. The study of _____, as contrasted with legal, obligations is called ethics
a) Behaviors b) Moral c) Business d) All of these
10. Ethics is a conception of _____conduct
a) Bad/Wrong b) Good/Correct c) Right/True d) Right and Wrong
11. Ethics and _____go together
a) Profits b) Management c) Worker ship d) All of these
12. _____driven companies are always successful in the long run
a) Management b) Worker union c) Value d) Profit
13. A business needs to be ethical, in order to
a) Fulfill public expectations for business
b) Prevent harming others
c) Improve business relations and employees productivity
d) All of these
14. Following is (are) types(s) of ethics issues(s)
a) Functional area ethics b) Corporate policy ethics
c) Face to face ethics d) All of these
15. What is called living with assumption for oneself as body and living of human being only on the basis of physical facilities, and not with right understanding and relationship?
a) Human consciousness b) Happiness
c) Right Understanding d) Animal Consciousness
16. Samridhi means
a) Happiness b) Wealth c) Prosperity d) Health
17. Developed nations are the live example of
a) Prosperity b) Wealth c) Happiness d) Health
18. The participation of human beings is seen in two forms
a) Prosperity and work b) Values and Understanding
c) Behavior and Wealth d) Behavior and Work
19. What are the outcomes of realization and understanding?
a) Work b) Values c) Happiness d) Health
20. Corporate culture _____ refers to how widely and deeply employees hold the company’s dominant values and assumptions
a) Breadth b) Depth c) Character d) Strength
21. The observable symbols and signs of a organization’s culture are called :
a) Its cultural design b) Artifacts c) Cultural formatting d) Its cultural dcor
22. _____is the feeling or responsibility and commitment for ensuring Right Understanding and Right Feeling in the self of my relative
a) Guidance b) Care c) Respect d) Affection
23. The ability to perform a specific task, action, or function successfully is called _____
a) Competence b) Purpose c) Intention d) Program

24. To live in harmony with other human beings is _____
a) Svatva b) Swatantrata c) Swarajya d) None of these
25. Natural acceptance is?
a) Universal b) Definite c) Continuous d) All of these
26. Basic human aspirations are _____
a) Happiness b) Prosperity c) Education d) Both (a) and (b)
27. Which of the following is true?
a) Happiness is definite and it can be understood
b) I need to be unhappy to recognize that I am happy
c) One will be bored of happiness if one is always happy
d) I will grow only if I am unhappy. If I become happy my growth will stop
28. To be wealthy is _____ condition in the modern world to be happy
a) Not essential b) True c) Fake d) The most important
29. It is important to have a good _____ to be successful in your career
a) Appear b) Attend c) Attitude d) Respectful
30. _____ skills such as writing, speaking and body language are important work ethics so that you can be clear in your messages to others
a) Communication b) Cooperation c) Honesty d) Integrity
31. You are a person with a high level of _____ if you stand by your values even if others put pressure on you to do something that is wrong or if they make fun of you
a) Communication b) Cooperation c) Honesty d) Integrity
32. If you don't have _____ you can really get in trouble for lying, cheating or deceiving others
a) Communication b) Cooperation c) Honesty d) Integrity
33. If you have _____ you will be a trusted person because they will see that you are committed to your company
a) Loyalty b) Organizational Skills c) Productivity d) Respect
34. Some jobs cannot be done alone and some can be done much better and faster if you possess the quality of _____
a) Responsibility b) Respect c) Teamwork d) Cooperation
35. The three major types of ethical issues include except?
a) Communication issues b) Systematic issues
c) Corporate issues d) Individual issues
36. Ethics are moral principles and values which _____
a) Guide a firm's behavior
b) Govern the actions of an individual
c) Provide employees with rules on how to behave
d) Are legally enforceable
37. The word "Ethics" comes from
a) Ethos b) Eternity c) Elementary d) Essentiality
38. A _____ is a set of principles that people hold themselves to or that are held by organizations or groups
a) Code of ethics b) Code of honor c) Ethical perspective d) Persuasive tactic

39. _____ are beliefs and attitudes we have that can actually conflict with our ethical decision
a) Morals b) Principles c) Virtues d) Values
40. Intellectual property rights are an important aspect in international trade. Which of the following is covered under IPR?
a) Patents b) Copyrights c) Trademarks d) All of these
41. Which of the following is not the infringement of copyright?
a) Copy the software to another computer by educational institution
b) Copy the software to another computer by company
c) Copy the software to another computer
d) To make backup copies
42. Patent can be granted for _____?
a) Process b) Ideas
c) Both product and process d) Product
43. Government may order the non advertisement of any patent application in the case of invention related to _____
a) Information technology b) Food
c) Drug d) Defense technology
44. Patent for the _____ is prohibited
a) Traditional knowledge b) Arms
c) Atomic energy d) Junk food
45. A design can be also registered under _____
a) Trademark Act b) Patent Act
c) Geographical Indication Act d) Copyright Act
46. IPR Complete specification gives _____
a) Full description of claim only
b) Full description of royalty
c) Full description of invention
d) Full description of application
47. Is the first owner of a Book's copyright?
a) Author b) Publisher c) Translator d) Printing press owner
48. Copyright is a _____
a) Creation by person of efforts, intellectual and capital
b) Creation by person of the efforts only
c) Creation by negligence of a person
d) Creation by the mental status of a person
49. Globalization has improved the living structure of which of the following?
a) All the people
b) People living in developing countries
c) People living in developed countries
d) None of these
50. Which sector experienced the least profit due to the effect of Globalization?
a) Industrial sector b) Service sector c) Agriculture Sector d) Secondary sector

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21CS61

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Software Engineering and Project Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What are seven broad categories of computer software? (10 Marks)
- b. What are the elements of a software process, generic process framework? (10 Marks)

OR

- 2 a. List the types of prescriptive process models and explain waterfall model and incremental process models. (10 Marks)
- b. Explain spiral model and concurrent models. (10 Marks)

Module-2

- 3 a. Define requirements engineering and explain seven different tasks of requirements engineering. (10 Marks)
- b. Draw and explain UMC use case diagram for SafeHome home security function. (10 Marks)

OR

- 4 a. Identify different ways of requirements model. Explain scenario based. (10 Marks)
- b. Draw and explain activity diagram for access camera surveillance via the internet display camera views function. (10 Marks)

Module-3

- 5 a. Explain 12 agility principles to achieve agility. (10 Marks)
- b. Explain SCRUM and FDD. (10 Marks)

OR

- 6 a. Explain Core Principles. (10 Marks)
- b. Explain Construction principles. (10 Marks)

Module-4

- 7 a. Explain activities covered in software project management. (10 Marks)
- b. Explain ways of categorizing software projects. (10 Marks)

OR

- 8 a. Explain Project charter. (10 Marks)
- b. Explain setting objectives. (10 Marks)

Module-5

- 9 a. Explain place of software quality in project planning. (10 Marks)
- b. Explain Boehm's model. (10 Marks)

OR

- 10 a. Explain V – Process Model. (10 Marks)
- b. Explain software Reliability. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

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21CS62

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Full Stack Development

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the processing of Django request and response with a flow diagram. (10 Marks)
- b. Write a python function in Django framework that displays date and time which is an offset of n hours from current date and time. (10 Marks)

OR

- 2 a. Illustrate the process of URL mapping with views. (10 Marks)
- b. Describe MVC design pattern with a neat diagram. (10 Marks)

Module-2

- 3 a. Define the following: (i) Tags (ii) Filters (10 Marks)
- b. Describe the process of template system usage in Django. (10 Marks)

OR

- 4 a. Explain the following:
(i) Selecting objects
(ii) Deleting objects (10 Marks)
- b. Summarize the concept of template loading in Django. (10 Marks)

Module-3

- 5 a. Illustrate the usage of admin interface in Django. (10 Marks)
- b. Develop a Model form for student that contains his topic chosen for project, languages used and duration with a model called project. (10 Marks)

OR

- 6 a. Describe the following:
(i) Creating a feedback form
(ii) Processing the submission (10 Marks)
- b. Create a Django app that performs student registration course. Register admin interfaces, perform migrations and illustrate data entry through admin forms. (10 Marks)

Module-4

- 7 a. Develop a Django app that performs CSV and pdf generation for any models created. (10 Marks)
- b. Illustrate the generic views of objects. (10 Marks)

OR

- 8 a. Create a generic class view which displays list of students and detail view that displays student details for any selected student in the list. (10 Marks)
- b. Explain the syndication feed framework in generating Non-HTML content. (10 Marks)

Module-5

- 9 a. Define the following: (i) HTML (ii) XML (iii) JSON (iv) CSS (10 Marks)
b. Mention the jQuery Ajax facilities and explain any four. (10 Marks)

OR

- 10 a. Illustrate the using of jQuery UI Autocomplete in Django templates. (10 Marks)
b. Explain XML HTTP Request by mentioning the properties. (10 Marks)

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Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Data Science and Visualization

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1
 - a. What is data science? Explain in detail the Venn diagram of data science. (08 Marks)
 - b. Explain the concept of datafication with an example. (04 Marks)
 - c. Explain the following concepts with examples:
 - i) Statistical Interference
 - ii) Population
 - iii) Samples
 - iv) Type of data. (08 Marks)

OR

- 2
 - a. Explain the role of a data scientist in the context of big data. How does the skill set of a data scientist differ from traditional data analysts? (08 Marks)
 - b. Explain the process of fitting a model to data. How do you evaluate the goodness of fit and what metrics are commonly used in evaluation? (08 Marks)
 - c. List out the common probability distributions with a brief explanation and examples with respect to usage in big-data. (04 Marks)

Module-2

- 3
 - a. Describe the data science process with neat diagram. (06 Marks)
 - b. Explain the K-means algorithm. List the issues associated with it. (06 Marks)
 - c. Explain the basic tools of exploratory data analysis with proper example. (08 Marks)

OR

- 4
 - a. Explain the concept of model evaluation in machine learning. What metrics are commonly used to evaluate the performance of linear regression, K-NN and K-means model? (10 Marks)
 - b. Discuss the importance of visualizing data in exploratory data analysis. What are the common tools and techniques used in EDA? (10 Marks)

Module-3

- 5
 - a. Define feature generation. Explain in detail how information can be categorized during feature generation. (08 Marks)
 - b. Explain and construct decision tree with a suitable example. (06 Marks)
 - c. What is dimensionality problem? Explain real world recommendation engine with neat diagram. (06 Marks)

OR

- 6 a. Explain singular value decomposition with a suitable illustration. (06 Marks)
 b. Explain the three primary methods used while building the regression model. (06 Marks)
 c. Write a short note on:
 i) Under fitting
 ii) Filters
 iii) Over fitting
 iv) Wrappers. (08 Marks)

Module-4

- 7 a. What is data wrangling with a neat diagram, explain the steps involved in data wrangling process. (08 Marks)
 b. Describe data visualization. Explain why data visualization is important. (06 Marks)
 c. Explain line chart, bar chart and radar chart with suitable examples. (06 Marks)

OR

- 8 a. Write a note on following:
 i) Scatter plot
 ii) Correlogram
 iii) Bubble plot
 iv) Heat map (08 Marks)
 b. Explain the stacked bar chart with an example. Explain the design practices to be followed while plotting stacked bar. (06 Marks)
 c. Explain the different map plots with a suitable example. (06 Marks)

Module-5

- 9 a. With a neat diagram, explain the components involved in anatomy of a matplotlib figure. (08 Marks)
 b. With proper illustration, explain how the bar chart and pie chart can be implemented using matplotlib. (06 Marks)
 c. Write a short notes on different customization options available while drawing any plot. (06 Marks)

OR

- 10 a. With a proper assumptions, explain how to use a scatter plot to visualize correlation between various animals. (06 Marks)
 b. Explain stacked area charts with a proper example plot the area chart using matplotlib. (06 Marks)
 c. Explain the basic operations that can be performed on image using matplotlib. (08 Marks)

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Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Data Mining and Data Warehousing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Datawarehouse. Explain the multi tier architecture of data warehouse. (10 Marks)
- b. Distinguish between OLTP and OLAP systems. (10 Marks)

OR

- 2 a. Define Data cube. Explain OLAP operations with neat diagram. (10 Marks)
- b. Explain the three schemas for multidimensional data models. (10 Marks)

Module-2

- 3 a. Define Data mining, with a neat diagram explain the KDD process. (10 Marks)
- b. What are the different strategies that are used in Data preprocessing? Explain Aggregation, sampling and dimensionality reduction with example. (10 Marks)

OR

- 4 a. Explain the challenges involved in Data mining. (10 Marks)
- b. Explain features subset selection, feature creation and discretization and binarization. (10 Marks)

Module-3

- 5 a. Write the pseudocode for the frequent itemset generation of the Apriori algorithm, explain. (10 Marks)
- b. List and explain the factor that affects the computational complexity of the Apriori algoirhtm. (10 Marks)

OR

- 6 a. Consider the transaction data set

IID	1	2	3	4	5	6
Items	{a, b}	{b, c, d}	{a, c, d, e}	{a, d, e}	{a, b, c}	{a, b, c, d}
7	8	9	10			
{a}	{a, b, c}	{a, b, d}	{b, c, d}			

Construct the FP tree by showing the trees separately after reading each transaction. Also find the frequent itemset ordered by their corresponding suffixes. (10 Marks)

- b. Generate frequent item sets and associated rules for the given transaction data set with minsup = 40% and minconf = 70% using Apriori algorithm. (10 Marks)

TID	Items bought
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Beer, Diapers}

Module-4

- 7 a. With an example, explain the different measures that can be used to determine the best way to split the records. (10 Marks)
- b. List and explain the different characteristics of decision tree induction. (10 Marks)

OR

- 8 a. Write and explain k-nearest neighbor classification, algorithm. Summarize the characteristics of nearest neighbor classifier. (10 Marks)
- b. Explain how a rule based classifier works? Also write the sequential covering algorithm to extract rules directly from data. (10 Marks)

Module-5

- 9 a. What is cluster analysis? Describe the different types of clustering techniques and clusters with example. (10 Marks)
- b. Explain DBSCAN algorithm for clustering with an example. (10 Marks)

OR

- 10 a. Explain basic Agglomerative Hierarchical clustering algorithm. Give its strength and weakness. (10 Marks)
- b. Write the CLIQUE algorithm. Also explain the strengths and limitations of CLIQUE. (10 Marks)

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21MAT655

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Advanced Linear Algebra

Time: 3 hrs.

Max. Marks: 100

Note : Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the vector space with examples. (06 Marks)
 b. Test the consistency and solve them $x - 3y + 4z = -4$, $3x + 7y + 7z = -8$ and $-4x + 6y - z = 7$. (07 Marks)
 c. Solve the linear system of equations $3x + 2y + 7z = 4$, $2x + 3y + z = 5$, $3x + 4y + z = 7$ using LU - decomposition. (07 Marks)

OR

- 2 a. Define the terms : i) Subspace ii) Bases and iii) Dimensions. (06 Marks)
 b. Prove that intersection of two subspaces of a V.S $V(F)$ is a subspace of $V(F)$. (07 Marks)
 c. Show that the set $S = \{V_1, V_2, V_3, V_4\}$ where $V_1 = (1, 0, 1, 0)$, $V_2 = (0, 1, -1, 2)$, $V_3 = (0, 2, 2, 1)$, $V_4 = (1, 0, 0, 1)$. (07 Marks)

Module-2

- 3 a. Define the terms : i) Linear transformation ii) Rank iii) Nullity. (06 Marks)
 b. Let v and w be V.S $V(F)$ and let T be a L.T from v into w and let v be a finite dimension. Prove that $\text{rank}(T) + \text{nullity}(T) = \dim v$. (07 Marks)
 c. Show that the mapping $T : R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + 1, y, z)$ are linear transformation. (07 Marks)

OR

- 4 a. Let v and w be V.S. $V(F)$ and let $T : v \rightarrow w$. Prove that T is invertible and T^{-1} is also invertible. (06 Marks)
 b. Show that the vectors $V_1 = (1, 1, 0, 0)$, $V_2 = (2, 0, 1, 0)$, $V_3 = (0, 1, 2, -1)$, $V_4 = (0, 1, -1, 0)$ forms a basis for R^4 . Find the coordinates of each of the standard basis vectors in the ordered basis $\{V_1, V_2, V_3, V_4\}$.. (07 Marks)
 c. Let $T : V \rightarrow W$ is a Linear transformations from the $V(F)$ to $W(F)$. Prove that
 i) $T(0) = 0$, ii) $T(-x) = -T(x)$ iii) $T(x - y) = T(x) - T(y)$. (07 Marks)

Module-3

- 5 a. Define i) Inner product space ii) Orthogonal sets iii) Orthogonal projections. (06 Marks)
 b. Let $u_1 = \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix}$, $u_2 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ and $y = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ prove that y as the sum of vector in w and a vector orthogonal to w . (07 Marks)

- c. Show that $\{V_1, V_2, V_3\}$ is an orthonormal basis of $\mathbb{R}^3$, where

$$V_1 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix}, \quad V_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad V_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix} \quad (07 \text{ Marks})$$

OR

- 6 a. Use Gram – Schmidt Orthogonalization process to vectors $(1, 0, 1)$, $(1, 0, -1)$ and $(0, 3, 4)$ to obtain an orthonormal basis $\{\alpha_1, \alpha_2, \alpha_3\}$ for $\mathbb{R}^3$. (10 Marks)
 b. Obtain the QR – decomposition for

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}. \quad (10 \text{ Marks})$$

Module-4

- 7 a. Find the Eigen values and Eigen vectors of $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$. (06 Marks)

- b. Obtain the Singular value decomposition of $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}$. (07 Marks)

- c. Diagonalize the following matrix, if possible $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$. (07 Marks)

OR

- 8 a. Explain the quadratic forms. (06 Marks)
 b. Explain the constrained optimization. (07 Marks)
 c. Explain the singular value decomposition. (07 Marks)

Module-5

- 9 a. Define the graphs and networks. (06 Marks)

- b. Draw the multigraph G whose adjacency matrix $M_A = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$. (07 Marks)

- c. Draw the undirected graphs K_3 and K_5 . (07 Marks)

OR

- 10 a. What are the applications of matrices in Engineering? (10 Marks)
 b. Describe the Computer Graphics. (10 Marks)

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Fourth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Design and Analysis of Algorithm

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Algorithm. Explain asymptotic notations Big O, Big Omega and Big theta notations with example. (10 Marks)
- b. Explain general plan of mathematical analysis of non-recursive algorithms with example. (10 Marks)

OR

- 2 a. Illustrate mathematical analysis of recursive algorithm for towers of hanoi. (08 Marks)
- b. Define time and space complexity. Explain important problem types. (12 Marks)

Module-2

- 3 a. Write the algorithm for recursive binary search and find efficiency for all three cases. (10 Marks)
- b. Explain divide and conquer technique. Write an algorithm for merge sort. (10 Marks)

OR

- 4 a. Illustrate the tracing of quick sort algorithm for the following set of numbers :
50, 10, 25, 30, 15, 70, 35, 55 (10 Marks)
- b. Explain decrease and conquer technique. Illustrate the topological sorting for the following graph : (10 Marks)

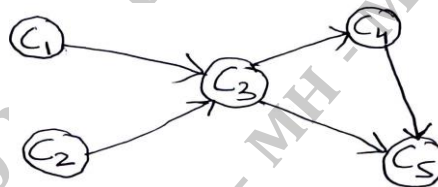


Fig. Q4 (b)

Module-3

- 5 a. Explain the concept of greedy method. Write a Kruskal's algorithm to find minimum cost spanning tree. (10 Marks)
- b. Solve the following single source shortest path problem assuming vertex '1' as the source. (10 Marks)

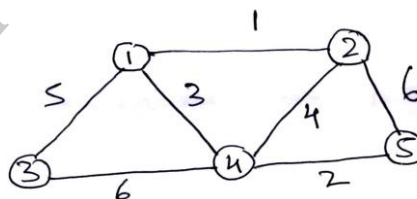


Fig.Q5 (b)

OR

- 6 a. Sort the given list of numbers using heap sort : 2, 9, 7, 6, 5, 8 (10 Marks)
 b. Construct a Huffman tree and resulting code word for the following :

Char	A	B	C	D	E	-
Probability	0.5	0.35	0.5	0.1	0.4	0.2

Encode the text DAD. Decode the text whose encode is 1100110110.

(10 Marks)

Module-4

- 7 a. Explain the concept of dynamic programming. Using Floy'ds algorithm. Solve the all pair shortest problem for the graph whose weight matrix is given below :

0	∞	3	∞
2	0	∞	∞
∞	7	0	1
6	∞	∞	0

(12 Marks)

- b. Write multistage graph algorithm to forward approach.

(08 Marks)

OR

- 8 a. Write an algorithm for Bellman-Ford algorithm. (10 Marks)
 b. Solve the following instance of Knapsack problem using dynamic programming. Knapsack capacity is 5. (10 Marks)

Item	1	2	3	4
Weight	2	1	3	2
Value	\$12	\$10	\$20	\$15

Module-5

- 9 a. Explain backtracking method. Illustrate 4-queens problem using backtracking method. And also write another solution. (10 Marks)
 b. Solve subset sum problem for the following example, $S = \{3, 5, 6, 7\}$ and $d = 15$. Construct a state space tree. (10 Marks)

OR

- 10 a. Explain branch and bound method. Solve assignment problem for the following :

	J_1	J_2	J_3	J_4
a	9	2	7	8
Person b	6	4	3	7
c	5	8	1	8
d	7	6	9	4

(10 Marks)

- b. With the help of a state space tree, solve following instance of the knapsack problem the FIFO branch and bound method. The knapsack capacity is 15.

Item	1	2	3	4
Weights	2	4	6	9
Values	10	10	12	18

(10 Marks)

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18CS43

Fourth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Operating Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain in detail about abstract view of the components of a computer system with a neat diagram. (10 Marks)
- b. Explain about computer system organization with a neat diagram. (10 Marks)

OR

- 2 a. Discuss briefly about operating system operations with diagram. (10 Marks)
- b. Discuss briefly about types of system calls with illustration. (10 Marks)

Module-2

- 3 a. Explain different types of multi threading modules with a neat diagram. (06 Marks)
- b. Explain Dining-philosopher's problem using monitors. (08 Marks)
- c. Explain various types of criteria for scheduling. (06 Marks)

OR

- 4 a. Explain three types of thread libraries. (06 Marks)
- b. Consider the following set of process. Draw Gantt charts and calculate average waiting time and average turn around time for i. non-preemptive SJF and preemptive SJF scheduling algorithms.

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

- c. Explain the Peterson's solution with its structure. (08 Marks)
- (06 Marks)

Module-3

- 5 a. What is a deadlock? What are the necessary conditions for the deadlock to occur? (05 Marks)
- b. How to prevent the occurrence of deadlock, explain in detail. (05 Marks)
- c. Consider the following snapshot of a system:

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	2	0	0	1	4	2	1	2	3	3	2	1
P ₁	3	1	2	1	5	2	5	2				
P ₂	2	1	0	3	2	3	1	6				
P ₃	1	3	1	2	1	4	2	4				
P ₄	1	4	3	2	3	6	6	5				

Answer the following using Banker's algorithm.

- i) Is the system in safe state? If so, give the safe sequence.
- ii) If process P₂ requests (0, 1, 1, 3) resources can it be granted immediately? (10 Marks)

OR

- 6 a. Explain paging hardware with TLB. (05 Marks)
b. Explain segmentation in detail. (05 Marks)
c. Discuss structure of page table with suitable diagrams. (10 Marks)

Module-4

- 7 a. Discuss briefly about demand – paging in memory management scheme. (10 Marks)
b. Discuss briefly about file attributes in a file system. (10 Marks)

OR

- 8 a. Explain in detail about various file operations in a file system. (10 Marks)
b. Explain in detail about various file types in a file system. (10 Marks)

Module-5

- 9 a. What is disk scheduling? Explain the following with a diagram:
i) FCFS ii) SSTF iii) SCAN. (10 Marks)
b. Write a short note on:
i) Linux file system ii) Linux process management. (10 Marks)

OR

- 10 a. Explain the several aspects of Disk management. (10 Marks)
b. Write a short note on:
i) Components of Linux system
ii) Process scheduling in Linux system. (10 Marks)

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18CS44

Fourth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Microcontroller and Embedded Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Differentiate between RISC and CISC processors. (06 Marks)
- b. Explain ARM core data flow model, with neat diagram. (08 Marks)
- c. Explain ARM registers used under various modes. (06 Marks)

OR

- 2 a. Explain the architecture of a typical embedded device based in ARM core, with a neat diagram. (08 Marks)
- b. Explain the various fields in the current program status register. (06 Marks)
- c. Discuss the following with diagram :
 - i) Von Neuman architecture with cache.
 - ii) Harvard architecture with TCM. (06 Marks)

Module-2

- 3 a. Explain Barrel Shifter Operation, with neat diagram. (06 Marks)
- b. Write an ALP using instruction to find out the factorial of a given number. (07 Marks)
- c. Write a program to add an array of 16 bit numbers and store the 32 bit result in internal RAM using ARM instructions. (07 Marks)

OR

- 4 a. Discuss the Load store instructions with respect to single register transfer along with various addressing modes. (10 Marks)
- b. Write an ALP program to multiply two 16 bit numbers. (05 Marks)
- c. With example, explain the Swap Instructions. (05 Marks)

Module-3

- 5 a. Differentiate between :
 - i) Microprocessor and Microcontroller. (08 Marks)
 - ii) Little Endian and Big Endian architecture. (06 Marks)
- b. With neat block diagram, explain the elements of embedded system. (06 Marks)
- c. Mention the application of embedded system with example of each. (06 Marks)

OR

- 6 a. Explain the different On board communication interfaces in brief. (08 Marks)
- b. Write a note on :
 - i) Reset circuit
 - ii) Watch dog timer. (06 Marks)
- c. Explain how program memory are classified. (06 Marks)

Module-4

- 7 a. Explain the Operational and non operational attributes of an embedded systems. (10 Marks)
b. Explain the different 'Embedded firmware design' approach in detail. (10 Marks)

OR

- 8 a. With a neat block diagram, explain design and working of Washing Machine. (10 Marks)
b. With FSM model, explain the design and operation of automatic tea/coffee vending machine. (06 Marks)
c. Explain Super loop based approach of embedded firmware design. (04 Marks)

Module-5

- 9 a. Define Process. Explain in detail the Structure of Memory Organisation and State transitions of the process. (10 Marks)
b. Explain Multithreading , Multiprocessing and Multitasking. (10 Marks)

OR

- 10 a. Explain with neat diagram, the concept of 'Dead lock' and mention the different conditions which favour a dead lock situation. (10 Marks)
b. With neat diagram, explain i) Binary Semaphore ii) Counting Semaphore. (10 Marks)

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Fourth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Object Oriented Concepts

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Write the difference between procedure oriented program and object oriented program. (04 Marks)
- b. Explain the various features of OOC. (08 Marks)
- c. What is constructor? List the different types of constructors and explain. (08 Marks)

OR

- 2 a. What is function overloading? Write C++ program to define three overload function area() to find area of circle, triangle and rectangle. (10 Marks)
- b. Why friend function is required? Write a program to add two numbers using friend function. (10 Marks)

Module-2

- 3 a. Discuss the label break and continue with example. (05 Marks)
- b. Explain the concepts of array in java with example. (07 Marks)
- c. List and explain Java Buzz words. (08 Marks)

OR

- 4 a. Explain different access specifiers in java with example. (10 Marks)
- b. What is nested class? Explain how nested class can be defined as private of enclosing class with example. (10 Marks)

Module-3

- 5 a. What is an exception? With syntax explain exception handling mechanism. (10 Marks)
- b. Define package. What are the steps involved in creating user defined package with an example. (10 Marks)

OR

- 6 a. Compare construct method overloading and overriding with example. (10 Marks)
- b. What are the uses of inheritance? Explain inheriting data members and method with a program. (10 Marks)

Module-4

- 7 a. What is thread? Explain two ways of creating a thread in JAVA with eg. (10 Marks)
- b. Explain synchronization using synchronized methods. (10 Marks)

OR

- 8 a. What is an applet? Explain the skeleton of an applet. Enlist applet tags. (10 Marks)
- b. What is meant by deadlock? How to avoid deadlock? Give example. (10 Marks)

Module-5

- 9 a. Explain the HTML applet with syntax and example. (10 Marks)
b. Explain advantages of swing over AWT and two key features of swings. (10 Marks)

OR

- 10 a. Explain a simple swing application with program. (10 Marks)
b. Explain with syntax:
i) JScrollPane
ii) Jlist
iii) ImageIcon
iv) JTextField
v) JtabbedPane. (10 Marks)

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Fourth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Data Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Data Communication. Explain the fundamental components of data communication. (05 Marks)
- b. With neat diagram explain the functionalities of each layer of OSI reference model. (10 Marks)
- c. What are ISPs? List and discuss different types of ISPs. (05 Marks)

OR

- 2 a. State the following terms : analog, digital, periodic, and non-periodic signals. Also sketch these signals. (06 Marks)
- b. What do you mean by transmission impairment? Explain the three causes of transmission impairment. (08 Marks)
- c. Write a note on the characteristics influencing the performance of the networks. (06 Marks)

Module-2

- 3 a. List any three line coding techniques, and represent the sequence 10110011 using techniques. (05 Marks)
- b. Explain the PCM techniques of changing analog signal to digital signal with neat diagram of PCM encoder and decoder. (10 Marks)
- c. What do you mean by sampling? Explain the three sampling methods with neat diagram. (05 Marks)

OR

- 4 a. Discuss the following transmission modes :
 - i) Parallel
 - ii) Serial
 - iii) Synchronous
 - iv) Asynchronous transmissions. (06 Marks)
- b. Define digital-to-analog conversion. List and define different types of digital-to-analog conversion. (06 Marks)
- c. Define frequency shift keying. Explain binary frequency shift keying and its implementation. (08 Marks)

Module-3

- 5 a. Highlight the concepts of multiplexing and list the categories of multiplexing. (04 Marks)
- b. Explain in detail synchronous time division multiplexing. (08 Marks)
- c. What is frequency hopping spread spectrum? Explain how it achieves bandwidth multiplexing. (08 Marks)

OR

- 6 a. Analyze in detail circuit switched network with neat diagram. (06 Marks)
 b. Describe virtual-circuit network. Discuss the five characteristics of virtual-circuit network. (08 Marks)
 c. Recall cyclic redundancy check with block diagram. Also explain CRC with an example. (06 Marks)

Module-4

- 7 a. What is the need for bit and byte stuffing at data link layer? Explain them with examples. (08 Marks)
 b. Discuss the three types of HDLC frames with neat diagrams. (08 Marks)
 c. Explain the transition phases of point-to-point protocol with relevant diagram. (04 Marks)

OR

- 8 a. What is channelization? List and explain the channelization protocols. (10 Marks)
 b. Discuss IPv4 addressing scheme and Dynamic Host Configuration Protocol (DHCP). (06 Marks)
 c. Write short notes on Networks Address Translation. (04 Marks)

Module-5

- 9 a. With a neat diagram, explain the structure of 802.3 frame format. (08 Marks)
 b. How Bridge Networks Improves DLL performance? What are the goals of Fast Ethernet? (08 Marks)
 c. Explain 10 Gigabit Ethernet implementation. (04 Marks)

OR

- 10 a. Describe the MAC layer in IEEE 802.11 standard. (06 Marks)
 b. Highlight the architecture of Bluetooth and explain its frame. (04 Marks)
 c. Explain the operation of cellular telephony with neat diagram. (10 Marks)

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18CS61

Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025 System Software and Compilers

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define System Software. Distinguish between system software and application software. (06 Marks)
b. Explain SIC/XE architecture (08 Marks)
c. Write an algorithm for pass 2 assembler of SIC assembler. (06 Marks)

OR

- 2 a. Explain the data structure and pass 1 algorithm of SIC assembler. (08 Marks)
b. List all assembler independent and dependent features and explain program relocation. (06 Marks)
c. What is loader? What are the basic functions the loader has to perform? (06 Marks)

Module-2

- 3 a. What is compiler? Explain various phases of compiler with the help of neat diagram. (10 Marks)
b. Explain the concept of input buffering with sentinels in the lexical analysis. (06 Marks)
c. List the formal definitions of operations on languages with notations. (04 Marks)

OR

- 4 a. Write the regular definition using extended regular expression notation and also draw the transition diagram to recognize the following tokens:
(i) Identifier (ii) Unsigned (10 Marks)
b. Explain three types of software productivity tools. (05 Marks)
c. Enlist algebraic laws for regular expressions. (05 Marks)

Module-3

- 5 a. Define left-recursion grammar, also write an algorithm to eliminate left recursion from a grammar. (05 Marks)
b. How to verify whether grammar is LL(1) or not? Show that :
$$S \rightarrow AaAb \mid BbBa$$
$$A \rightarrow E$$
$$B \rightarrow E$$
is LL(1), without constructing any table. (10 Marks)
c. For the grammar $A \rightarrow (A) \mid a$, construct LR(0) set of items (05 Marks)

OR

- 6 a. Explain the working of shift reduce parser. Parse the input string $id * id$ using the grammar.
- $$E \rightarrow E + T \mid T$$
- $$T \rightarrow T * F \mid F$$
- $$F \rightarrow (E) \mid id$$
- (08 Marks)
- b. With a diagram, explain the model of an LR parser. (04 Marks)
- c. For the given grammar $E \rightarrow E + n/n$ construct parsing table of LL(1). Verify $3+4+7$ and show each step of verification with reference to parsing table. (08 Marks)

Module-4

- 7 a. Explain the structure of LEX program, with an example. (06 Marks)
- b. Write a LEX program for the tokens given below:

Lexemes	Token Name	Attribute value
Any WS	-	-
if	if	-
then	then	-
else	else	-
Any id	id	Ptr to table entry
Any number	number	Ptr to table entry
<	relop	LT
<=	relop	LE
=	relop	EQ
< >	relop	NE
>	relop	GT
>=	relop	GE

- c. Write a LEX program to count the number of vowels and consonants in a given input string. (10 Marks)
- (04 Marks)

OR

- 8 a. List and explain with an example the different wildcard characters used in LEX. (08 Marks)
- b. Write a YACC program to evaluate the arithmetic expression. (06 Marks)
- c. Explain the structure of YACC program. (06 Marks)

Module-5

- 9 a. Write annotated parse tree for $3 * 5 + 4n$ using top down approach. Write semantic rules for each step. (10 Marks)
- b. Define (i) Synthesized attribute (ii) Inherited attribute. (06 Marks)
- c. Explain the concept of syntax directed definition. (04 Marks)

OR

- 10 a. Construct DAG and three address code for the following expression:
 $a + a * (b - c) + (b - c) * d$ (04 Marks)
- b. Explain the following with an example:
 i) quadruples ii) triples (08 Marks)
- c. Discuss the various issues in the design of a code generator. (08 Marks)

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Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Computer Graphics and Visualization

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is DDA? With the help of a suitable example demonstrate the working principle of Bresenham's Line drawing algorithm for different slopes of a line. (10 Marks)
- b. With a neat diagram, explain the basic design and operation of Cathode Ray tube. (10 Marks)

OR

- 2 a. Explain with diagram the different Cartesian reference frames are used in the process of constructing and displaying a scene. (10 Marks)
- b. With a neat diagram, explain the architecture of a raster display system with integrated display processor. (10 Marks)

Module-2

- 3 a. Give the reason to convert transformation matrix to homogeneous co-ordinate representation and show the process of conversion. Shear the polygon A(2, 2), B(4, 2), C(4, 4), D(3, 5), E(2, 4) along X-axis with a shearing factor of 0.2 (10 Marks)
- b. Explain General scan line polygon fill algorithm support your claim with a neat diagram. (10 Marks)

OR

- 4 a. Explain with example any two algorithms used to identify the interior area of a polygon. (10 Marks)
- b. Develop composite homogenous transformation matrix to rotate an object with respect to a pivot point. For the triangle A(3, 2), B(6, 2), C(6, 6) rotate it in anticlockwise direction by 90° degree keeping A(3, 2) fixed, draw the new polygon. (10 Marks)

Module-3

- 5 a. Explain basic illumination models. (10 Marks)
- b. Obtain the matrix representation for rotation of a object about an arbitrary axis. (10 Marks)

OR

- 6 a. Explain RGB and CMY color models with examples. Explain the transformation between CMY and RGB color spaces. (10 Marks)
- b. Explain Cohen Sutherland line clipping Algorithm clip. The lines with coordinates, $(X_0, Y_0) = (60, 20)$, $(X_1, Y_1) = (80, 120)$, given the window boundaries. $(X_{wmin}, Y_{wmin}) = (50, 50)$ and $(X_{wmax}, Y_{wmax}) = (100, 100)$ (10 Marks)

Module-4

- 7 a. Explain 2 classifications of visible surface detection algorithm. (05 Marks)
- b. Compare perspective and parallel projection. (05 Marks)
- c. What is 3D viewing? With the help of block diagram, explain 3D viewing pipeline architecture. (10 Marks)

OR

- 8 a. Design the transformation matrix for perspective projection and give OpenGL 3D viewing function. (10 Marks)
- b. Explain perspective projections with reference point and vanishing point with neat diagram. (10 Marks)

Module-5

- 9 a. List the properties of Bezier curve and also explain Bazier techniques of generating curves. (10 Marks)
- b. Explain Request, sample and event input modes with block diagram. (10 Marks)

OR

- 10 a. With the help of a suitable programming construct, explain event driven input menu picking and building interactive models. (10 Marks)
- b. With the role of glCallList() function in creating DisplayLists in OpenGL. Write OpenGL code for rendering a Simple Animated face. (10 Marks)

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Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025
Web Technology and its Applications

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1**
 - a. Explain the Anchor (a) tag to create wide range of links with suitable example. **(08 Marks)**
 - b. Discuss the advantages of semantic HTML elements. **(06 Marks)**
 - c. Write HTML code segment to display following context using list elements :
 1. Introduction
 2. Literature survey
 3. My solution - Methodology
 - Results
 4. Conclusion.**(06 Marks)**

OR

- 2 a. What is CSS? Explain different location of styles with suitable example. (06 Marks)
- b. What is Selector? Explain any 4 selector with suitable example. (06 Marks)
- c. Discuss the cascading principles with examples. (08 Marks)

Module-2

- 3 a.** Design the following table using HTML code :

Items	Details		Total Amount
	Units	Amount	
Computers	50000/-	2	100000
Projector	25000/-	3	75000
Total Amount			175000

- b. Explain basic form structure with suitable example. (06 Marks)
- c. Explain different types of button control elements. (06 Marks)

OR

4. a. Briefly explain any two positioning elements with suitable example. (08 Marks)
b. What is Micro format? Explain with neat diagram. (06 Marks)
c. Design 3 column layout using positioning and float property with suitable example. (06 Marks)

Module-3

- 5 a. What is client side scripting? Discuss its advantages and disadvantages. (06 Marks)
- b. List any four methods of DOM. Write the code segment to modify the context of HTML documents using DOM methods. (08 Marks)
- c. Write a Java Script code to calculate cubes of numbers from 0 to n. (06 Marks)

OR

Important Note :

1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, $42+8=50$, will be treated as malpractice.

- 6 a. List the keyboard events. Write Java script code segments to display key pressed in the keyboard on alert message. (08 Marks)
 b. List any six web server responsibilities. (06 Marks)
 c. Explain different types of parameters passing in PHP , with suitable example. (06 Marks)

Module-4

- 7 a. Discuss how arrays are different in PHP compare to other tradition programming languages. (06 Marks)
 b. Explain different file size restriction mechanism with suitable code segments. (08 Marks)
 c. Illustrate data flow from HTML form to PHP. (06 Marks)

OR

- 8 a. Implement a PHP code to create a class with following specification :
 Data members : E_name , E_No , DOB , Salary
 Member functions : Read , Write.
 Use the above specification to read and write the 2 employee information. (08 Marks)
 b. What is Interface? Discuss the implementation of Interface with suitable example. (06 Marks)
 c. Explain static member with suitable example. (06 Marks)

Module-5

- 9 a. What is Cookies? Write PHP code segment to read and write cookies. (06 Marks)
 b. Discuss how web storages are better than cookies. Write PHP code segment to read from webstorage. (08 Marks)
 c. What is Caching? Explain page output caching with neat diagrams. (06 Marks)

OR

- 10 a. Write PHP code segment to convert JSON string into PHP object. (06 Marks)
 b. Explain MVC frame work with neat diagram. (08 Marks)
 c. What is AJAX? Explain how the asynchronous request is handled with suitable diagram. (06 Marks)

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Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Data Mining and Data Warehousing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is data warehouse? Explain three tier architecture of data warehousing with the help of block diagram. (10 Marks)
- b. Explain different features of data warehousing. (05 Marks)
- c. Distinguish between OLTP and OLAP. (05 Marks)

OR

- 2 a. Illustrate all the conceptual modelling of data warehousing with an example for each. (10 Marks)
- b. Explain concept of hierarchy. (05 Marks)
- c. Identify different data cube measures and explain briefly. (05 Marks)

Module-2

- 3 a. Illustrate the data preprocessing steps in brief. (10 Marks)
- b. Discuss the difference between HOLAP, ROLAP, MOLAP with an example each. (10 Marks)

OR

- 4 a. Illustrate indexing of OLAP data with an example. (10 Marks)
- b. Discuss the challenges of data mining. (10 Marks)

Module-3

- 5 a. Illustrate frequent item set generation with an example by applying association rule. (10 Marks)
- b. Generate rules for the following transaction details using Apriori algorithm. Assume Minsup = 50%, MinConf = 70%.

TID	
1	{A, C, D}
2	{B, C, E}
3	{A, B, C, E}
4	{B, E}

(10 Marks)

OR

- 6 a. Illustrate Apriori algorithm for the frequent Itemset generation. (10 Marks)
 b. Explain the steps of FP – Growth algorithm. Construct FP – Tree for the below example.

FID	Item bought
100	{f, a, c, d, g, i, m, p}
200	{a, b, c, f, l, m, o}
300	{b, f, h, j, o}
400	{b, c, k, s, p}
500	{a, f, c, e, l, p, m, n}

MinSupp = 3.

(10 Marks)

Module-4

- 7 a. What is classification task? Illustrate with an example. (10 Marks)
 b. Discuss Hunt's algorithm. Construct decision tree induction using Hunt's algorithm for the following data :

TID	House owner	Marital status	Annual income	Defaulted borrower
1	Yes	Single	125 K	No
2	No	Married	100 K	No
3	No	Single	70 K	No
4	Yes	Divorced	95 K	Yes
5	No	Single	90 K	Yes
6	No	Married	60 K	No

(10 Marks)

OR

- 8 a. Illustrate Nearest Neighbor classifier with an example. (10 Marks)
 b. Explain sequential covering algorithm for rule based classifier. Demonstrate with an example. Assume any data set that must contain a collection of positive and negative values. (10 Marks)

Module-5

- 9 a. Illustrate K-means algorithm with an example. (10 Marks)
 b. Explain the method for cluster evaluation and density based clustering. (10 Marks)

OR

- 10 a. Illustrate graph based clustering with an example. (10 Marks)
 b. Discuss scalable clustering algorithm with an example. (10 Marks)

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18CS651

Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Mobile Application Development

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define android. Explain the architecture of android with the help of diagram. (10 Marks)
- b. Define intent. Demonstrate how to pass data between activities with intents using intent Extras. (10 Marks)

OR

- 2 a. With the help of flow diagram, describe different states of an activity in its life cycle. (10 Marks)
- b. With the help of code snippet. Demonstrate how to get back data from an activity and process it. (10 Marks)

Module-2

- 3 a. How to create raised button and flat button? Explain it, Also mention 3 states of each button. (10 Marks)
- b. With necessary code snippet, explain the steps for floating context menu. (10 Marks)

OR

- 4 a. Define recycler view. Demonstrate with code snippet, the usage of recycler view. (10 Marks)
- b. What is Testing? Explain manual testing and automated testing of an app. (10 Marks)

Module-3

- 5 a. With neat diagrams, explain the lifecycle of services. (08 Marks)
- b. Define notification. Explain the steps involved in creating and setting notifications. (07 Marks)
- c. Explain the 2 types of broadcast intent. (05 Marks)

OR

- 6 a. Define loader. Explain the callback methods of LoadManager class along with how to start and restart loader. (08 Marks)
- b. Explain the two ways of registering the broadcast receiver. (07 Marks)
- c. Explain prefetching with example. (05 Marks)

Module-4

- 7 a. Define shared preferences. Differentiate between shared preferences and saved instance state. (06 Marks)
- b. Define and list the characteristics of SQLite database. (08 Marks)
- c. Define Transaction. Explain ACID properties. (06 Marks)

OR

- 8 a. Define content provider. Explain with an example, sharing data between apps using content provider. (08 Marks)
- b. List and explain the steps of implementing an SQLite database. (12 Marks)

Module-5

- 9 a. Illustrate the best practices to keep user data secure. (08 Marks)
- b. Define permission. Explain it is needed along with its types. (06 Marks)
- c. Write a note on AdMob. (06 Marks)

OR

- 10 a. Define APK. Explain the steps involved in preparing App for releasing. (10 Marks)
- b. With an example, explain firebase real time database and also describe how the data is stored in it. (10 Marks)

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Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Programming in Java

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain any four features of OOP. (08 Marks)
- b. What is literal? Explain different types of literals in Java. (06 Marks)
- c. What is Scope of variable in Java? Explain with an example program. (06 Marks)

OR

- 2 a. Explain control statements in Java with example program. (08 Marks)
- b. What is an array? Explain how 1D and 2D arrays are declared in Java. Write a program to find sum of all the elements of a two dimensional array. (08 Marks)
- c. Write a Java Program to find area and perimeter of a circle. (04 Marks)

Module-2

- 3 a. Describe bitwise operators with example. (08 Marks)
- b. Explain different access modifiers in Java. (04 Marks)
- c. Write a program to implement simple calculator using,
(i) if-else
(ii) Switch statements (08 Marks)

OR

- 4 a. Explain iterative statements in Java with an example for each. (08 Marks)
- b. Write a Java program to search an element and find its position in an array by using linear search. (06 Marks)
- c. Write Java program to find largest of three numbers using, (i) nested if (ii) ternary operator. (06 Marks)

Module-3

- 5 a. What is constructor? Explain different types of constructors with example. (06 Marks)
- b. Explain the operation of stack with example program. (08 Marks)
- c. What is inheritance? Explain single inheritance with example. (06 Marks)

OR

- 6 a. What is overloading of methods? Explain with example program. (05 Marks)
- b. Distinguish between call by value and call by reference with example. (08 Marks)

- 6 c. Write an abstract class shape, which has an abstract method area (). Derive three classes Triangle, Rectangle and Circle from the shape class and to override area (). Implement run time polymorphism by creating array of references to super class. Compute area of different shapes and display the same. (07 Marks)

Module-4

- 7 a. What is Package in Java? Explain with an example. (06 Marks)
b. Distinguish between abstract classes and abstract methods. (06 Marks)
c. What is nested try statement? Write program to illustrate the same. (08 Marks)

OR

- 8 a. What is exception handling in Java? Explain with an example, the use of try and catch. (06 Marks)
b. Explain interface, with an example program illustrate the implementation of an interface. (08 Marks)
c. Explain the operation of throw with a program. (06 Marks)

Module-5

- 9 a. What is enumeration in Java? Explain with example program. (07 Marks)
b. Write a program to read data from file. (07 Marks)
c. Explain different type wrappers in Java. (06 Marks)

OR

- 10 a. What is Java applet? Explain program to display "Hello" using applet. (08 Marks)
b. Explain special string operations. (08 Marks)
c. What are transient and volatile modifiers? (04 Marks)

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Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025

File Structures

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the functions OPEN, READ, WRITE, SEEK with parameters. (08 Marks)
- b. Explain the sector based data organization in magnetic disk with a neat diagram. (08 Marks)
- c. Explain the concept of inheritance using the IP buffer class hierarchy. (04 Marks)

OR

- 2 a. Define field and record. Explain the different methods for organizing fields and records of a file, with examples. (12 Marks)
- b. Define: i) RRN ii) Physical file iii) Logical file iv) File access and file organization. (08 Marks)

Module-2

- 3 a. Describe the operations required to maintain an indexed file. (08 Marks)
- b. Build the Huffman tree and code the input symbols for the following input sequence cddaf.

Symbol	a	b	c	d	e	f
Probability	0.1	0.4	0.06	0.1	0.04	0.3

- c. Explain the limitations of binary searching and internal sorting. (08 Marks)
- (04 Marks)

OR

- 4 a. Explain the different way of reclaiming space in files. (10 Marks)
- b. Discuss the limitations of retrieving the records using combination of secondary key. Explain the solution by using linking list of reference techniques. (10 Marks)

Module-3

- 5 a. Write an algorithm for heap sorting method. Show the construction of heap tree for following incoming sequence FDCGHIBEA. (10 Marks)
- b. What is co-sequential processing and what are the assumptions and components of the model? (10 Marks)

OR

- 6 a. What is multilevel indexing? Explain the concept of B-Trees in multilevel indexing. Construct a B-tree for the sequence (order 4) Q, P, A, E, C, X, Z, D, H, I, F. (10 Marks)
- b. With examples explain the following operations in a B-tree: (10 Marks)
- i) Deletion ii) Merging iii) Redistribution.

Module-4

- 7 a. Give the structure of indexed sequential access with a neat diagram. Discuss simple prefix B+ tree maintenance. (10 Marks)
- b. Compare : B trees, B+ trees, Simple prefix, B+ trees. (10 Marks)

OR

- 8 a. Explain a B tree and its creation with example. (10 Marks)
b. Explain the internal structure of index set blocks. (10 Marks)

Module-5

- 9 a. Discuss any 3 methods used to avoid collision in hashing. (10 Marks)
b. Suppose that 10000 addresses are allocated to hold 8000 records in a randomly hashed file and that each address can held one record compute the following values.
i) The packing density
ii) The expected number of address with no records assigned.
iii) The expected number of address with one record assigned.
iv) Expected number of overflow records. (10 Marks)

OR

- 10 a. Explain :
i) Dynamic Hashing
ii) Linear Hashing
iii) Extendible Hashing. (12 Marks)
b. Explain the working of extendible hashing. (08 Marks)

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CBGS SCHEME

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18IS62

Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Software Testing

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain program behavior insights from a Venn Diagram for functional testing and structural testing. (10 Marks)
- b. Identify and explain fault taxonomies with example. (10 Marks)

OR

- 2 a. With the flowchart for the traditional triangle problem implementation. (10 Marks)
- b. Analyse and explain the SATM screen. (10 Marks)

Module-2

- 3 a. Explain the process of Boundary value analysis in detail with example. (10 Marks)
- b. Develop test case using robust worst BVA testing for triangle problem. (10 Marks)

OR

- 4 a. Explain the variants of equivalence class testing. Derive equivalence class test case for next date problem. (10 Marks)
- b. Briefly explain Mutation Analysis step. (05 Marks)
- c. Write note on Mutation Analysis (05 Marks)

Module-3

- 5 a. Analyze and explain metric – based testing. (10 Marks)
- b. Explain define/Use testing with example. (10 Marks)

OR

- 6 a. Describe about scaffolding. Discuss about Generic versus specific scaffolding. (08 Marks)
- b. Define :
 - i) Test oracles
 - ii) Self – checks
 - iii) Capture
 - iv) Replay.(12 Marks)

Module-4

- 7 a. Explain about the following basic principle of Testing process framework.
i) Sensitivity
ii) Restriction. (08 Marks)
- b. What are dependability properties in testing process framework? Explain with diagram. (08 Marks)
- c. Write short notes on Test design specification document. (04 Marks)

OR

- 8 a. Explain in detail about the Risk management interms of process and quality management. List out various Risks and their control tactics in both. (10 Marks)
- b. Write short notes on the following :
i) Organizing documents
ii) Test and analysis Reports. (10 Marks)

Module-5

- 9 a. Analyze and explain integration testing strategies. (10 Marks)
- b. What is regression testing? Explain regression test selection technique. (10 Marks)

OR

- 10 a. Explain Rapid Prototyping Life Cycle with diagram. (10 Marks)
- b. Explain Decomposition – Based Integration. (10 Marks)

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CBCS SCHEME

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18MAT41

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Complex Analysis, Probability and Statistical Methods

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Show that $w = f(z) = z + e^z$ is analytic and hence find $\frac{dw}{dz}$. (06 Marks)
- b. Derive Cauchy's – Riemann equations in polar form. (07 Marks)
- c. If $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ then find analytic function $f(z) = u + iv$. (07 Marks)

OR

- 2 a. Show that the real and imaginary parts of an analytic function $f(z) = u + iv$ are harmonic. (06 Marks)
- b. If $f(z)$ is an analytic function then show that $\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] |f(z)|^2 = 4 |f'(z)|^2$. (07 Marks)
- c. If $u = \left[r + \frac{1}{r} \right] \cos \theta$ then find the corresponding analytic function $f(z) = u + iv$. (07 Marks)

Module-2

- 3 a. State and prove Cauchy's integral formula. (06 Marks)
- b. Discuss the conformal transformation $w = f(z) = z^2$. (07 Marks)
- c. Find the bilinear transformation which maps the points $z = 1, i, -1$ into the points $w = i, 0, -i$. (07 Marks)

OR

- 4 a. Evaluate $\int_C z^2 dz$ along the curve made up of two line segments, one from $z = 0$ to $z = 3$ and another from $z = 3$ to $z = 3 + i$. (06 Marks)
- b. Evaluate $\int_C \frac{e^{2z}}{(z+1)(z-2)} dz$, where C is the circle $|z| = 3$. (07 Marks)
- c. Find the bilinear transformation which maps the points $z = -1, 0, 1$ into the points $w = 0, i, 3i$. (07 Marks)

Module-3

- 5 a. The probability distribution of a random variable X is given by the following table:

$X(=x_i)$	-3	-2	-1	0	1	2	3
$P(X)$	k	2k	3k	4k	3k	2k	k

Find (i) The value of k , (ii) $P(x \leq 1)$, (iii) $P(-1 < x \leq 2)$

(06 Marks)

- b. The probability that a pen manufactured by a factory be defective is $1/10$. If 12 such pens are manufactured, what is the probability that (i) Exactly 2 are defective (ii) Atleast 2 are defective (iii) None of them are defective. (07 Marks)
- c. The length of telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that a random call made from this booth (i) Ends less than 5 minutes (ii) Between 5 and 10 minutes. (07 Marks)

OR

- 6 a. The probability density function of a random variable X is

$$f(x) = \begin{cases} Kx^2 & , 0 < x < 3 \\ 0 & , \text{otherwise} \end{cases}$$

Find (i) The value of K (ii) $P(1 < x < 2)$ (iii) $P(x \leq 1)$ (06 Marks)

- b. In a certain town the duration of a shower is exponentially distributed with mean 5 minutes what is the probability that a shower will last for (i) Ten minutes or more (ii) Less than Ten minutes (iii) Between 10 and 12 minutes. (07 Marks)
- c. The marks of 1000 students in an examination follows a normal distribution with mean 70 and standard deviation 5. Find the number of students whose marks will be (i) less than 65, (ii) more than 75 (iii) 65 to 75. [$\phi(1) = 0.3413$] (07 Marks)

Module-4

- 7 a. Compute the rank correlation coefficient for the following data:

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

(06 Marks)

- b. Find a best fitting straight line $y = ax + b$ for the data below:

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

(07 Marks)

- c. Obtain the lines of regression and hence find the coefficient of correlation for the data below:

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

(07 Marks)

OR

- 8 a. If θ is the acute angle between the lines of regression then show that

$$\tan \theta = \frac{\sigma_x \sigma_y \sqrt{1 - r^2}}{\sigma_x^2 + \sigma_y^2 \left[\frac{r}{\sigma_x} \right]}$$

(06 Marks)

- b. Find a best fitting second degree parabola of the form $y = ax^2 + bx + c$ for the data below:

x	1	2	3	4	5
y	10	12	13	16	19

(07 Marks)

- c. Find the coefficient of correlation for the following data:

x	10	14	18	22	26	30
y	18	12	24	06	30	36

(07 Marks)

Module-5

- 9 a. The joint probability of discrete random variables X and Y is given below:

	Y	1	3	9
X				
2		1/8	1/24	1/12
4		1/4	1/4	0
6		1/8	1/24	1/12

Determine (i) Marginal distribution of X and Y. (ii) Covariance and correlation of X and Y.

(06 Marks)

- b. A survey was conducted in a slum locality of 2000 families by selecting a sample of size 800, it was revealed that 180 families were illiterates. Find the probable limits of the illiterates families in the population of 2000 at 1% level of significance. (07 Marks)
- c. A group of 10 boys fed at diet A and another group of 08 boys fed on another diet B for a period of 06 months record the following increase in weights in pounds.

Diet A	05	06	08	01	12	04	03	09	06	10
Diet B	02	03	06	08	10	01	02	08	-	-

Test whether diet A and B differ significantly regarding their effect on increase in weight. [$t_{0.05} = 2.12$]

(07 Marks)

OR

- 10 a. Explain the terms :
 (i) Null hypothesis
 (ii) Type-I and Type-II errors
 (iii) Level of significance. (06 Marks)
- b. A certain stimulus administered to each of the 12 patients resulted in the following change in blood pressure: 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4
 Can it be concluded that the stimulus will increase the blood pressure? [$t_{0.05} = 2.201$] (07 Marks)
- c. A sample analysis of examination, result of 500 students was made, it was found that 220 students had failed, 170 had secured third class, 90 had secured second class, 20 had secured first class. Do these figures support the general examination result which is in the ratio 4 : 3 : 2 : 1 for the respective categories? [$\chi^2_{0.05} = 7.81$]. (07 Marks)

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